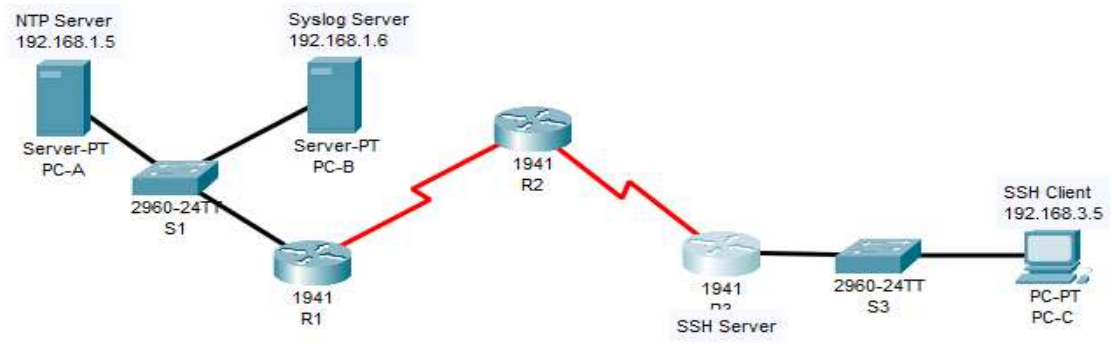
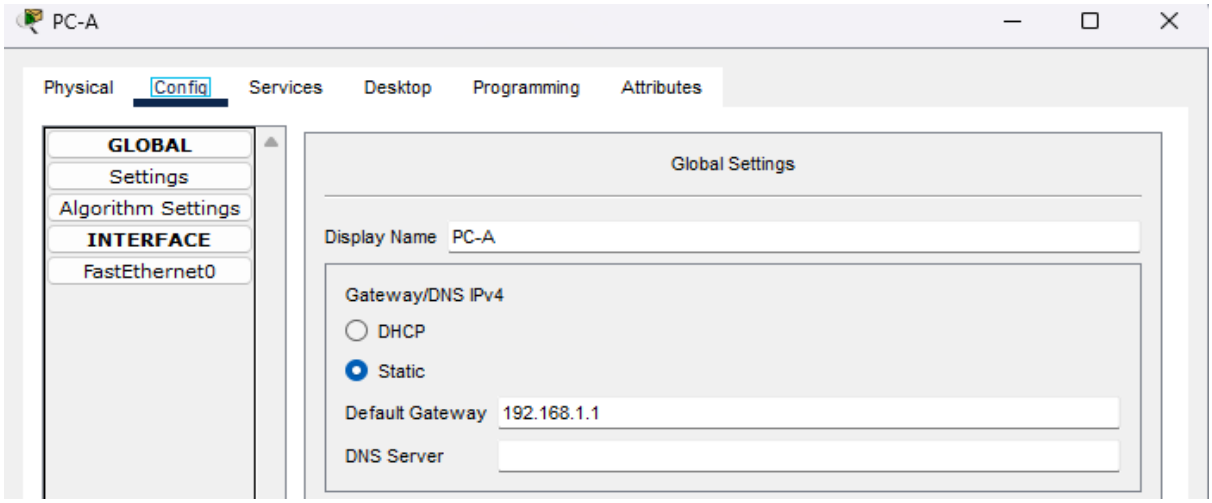
Practical - 1

Configure Cisco Routers for Syslog, NTP, and SSH Operations Topology

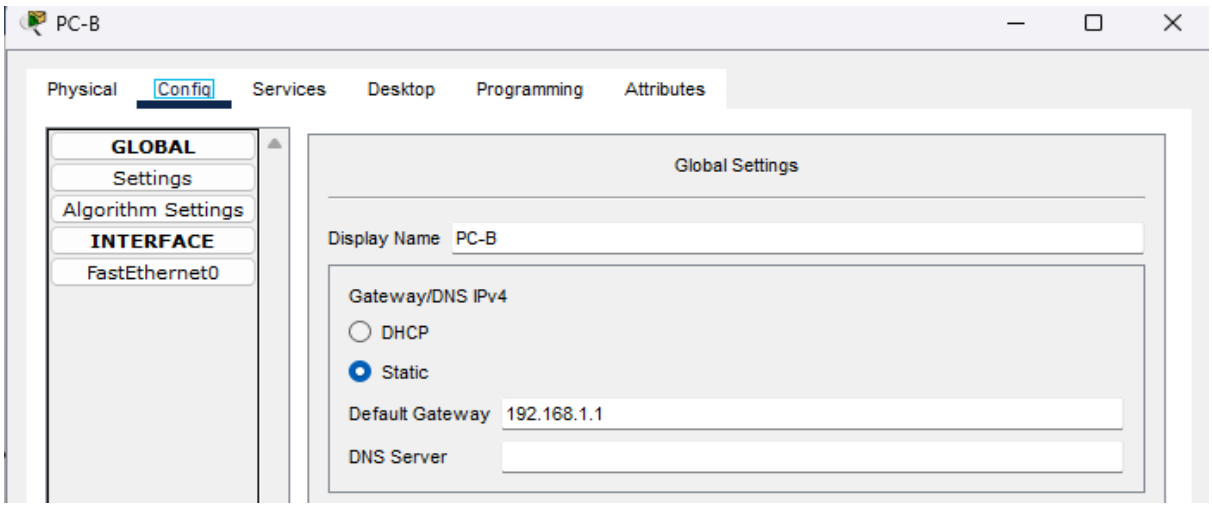


Configurations:

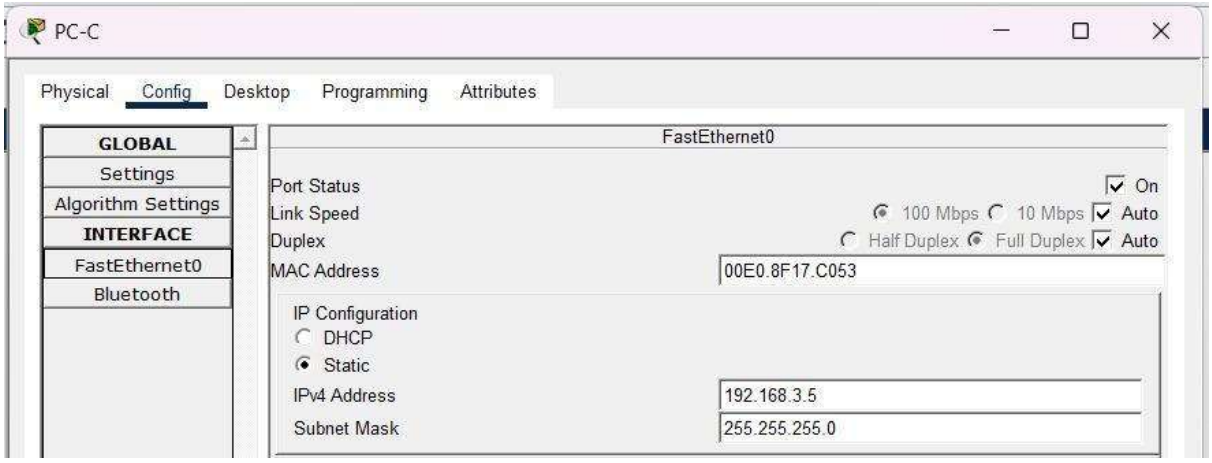
Server1



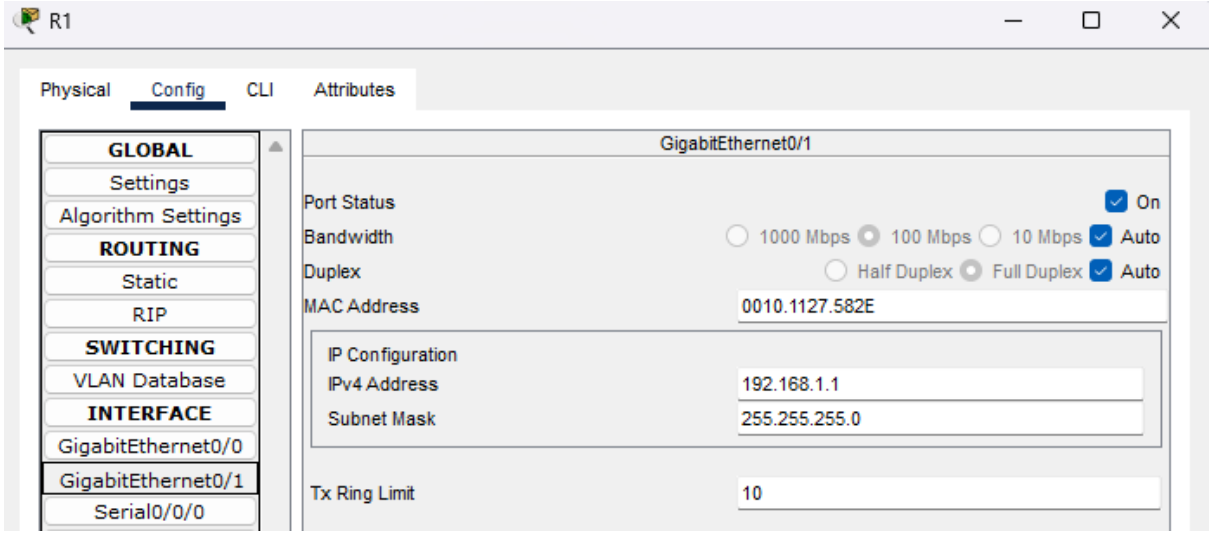
Server 2

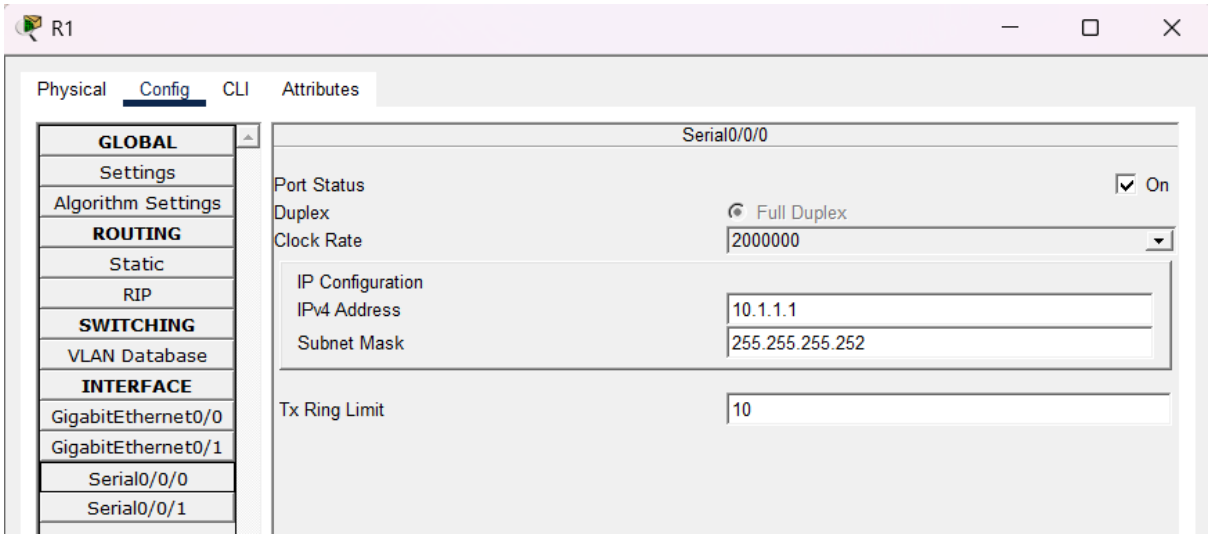


PC-C

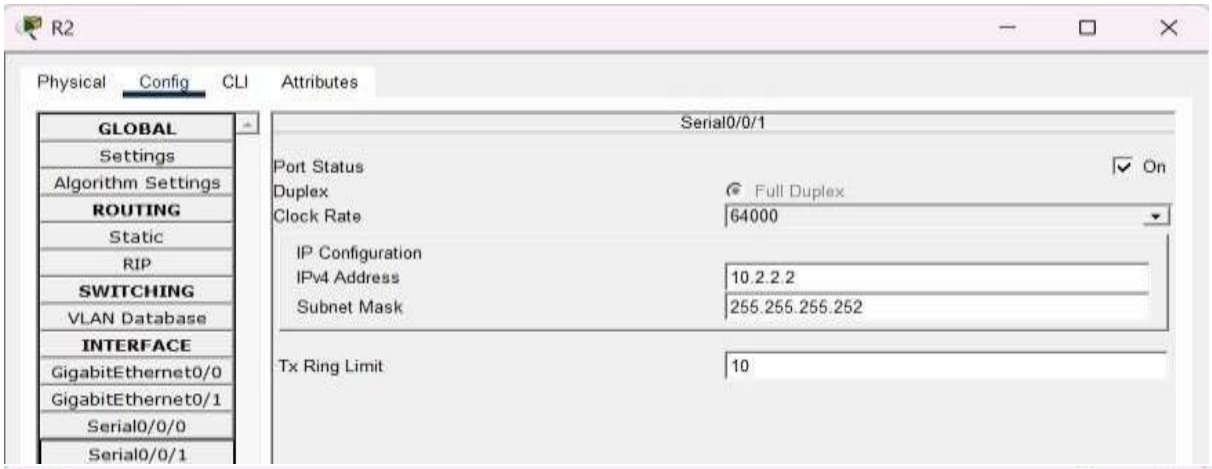

PC-C
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.8F17.C053
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.5
Subnet Mask
255.255.255.0

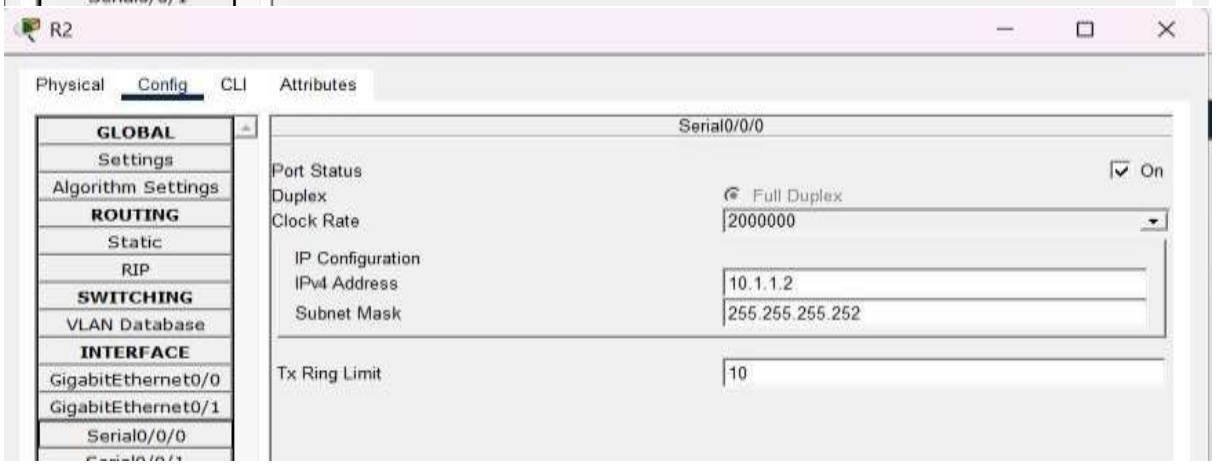
R1


R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0010.1127.582E
IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

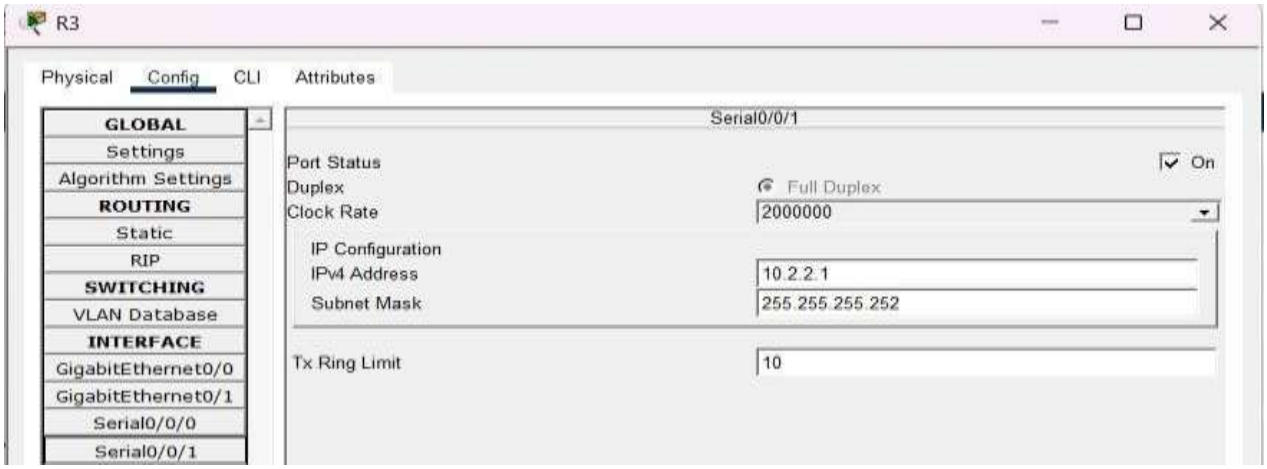

R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

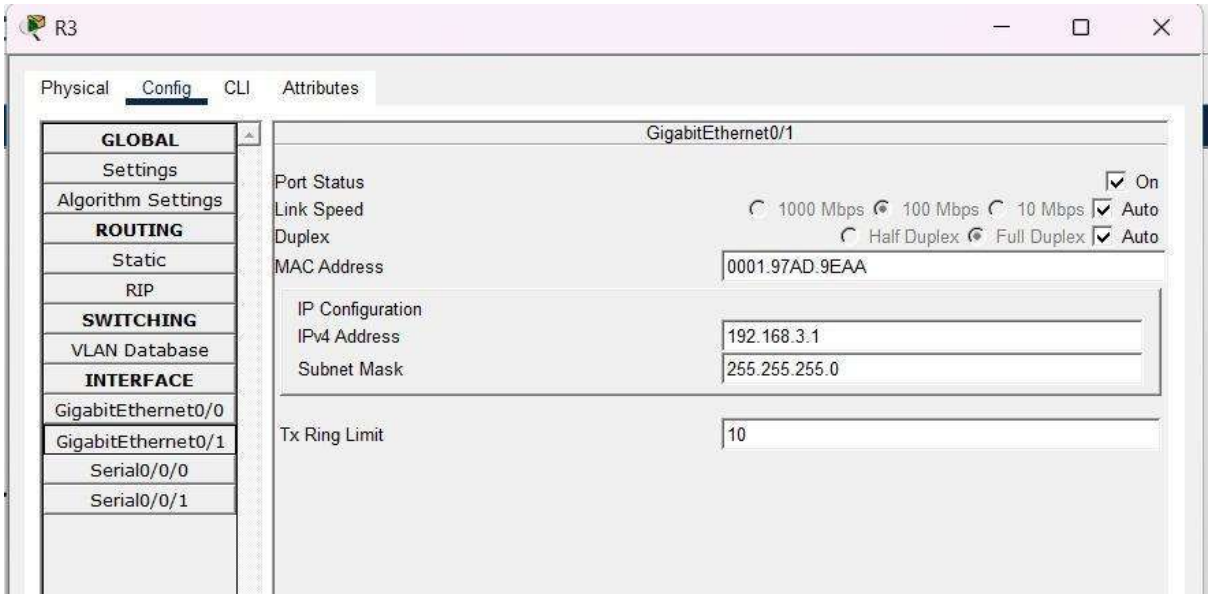
R2


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
64000
IP Configuration
IPv4 Address
10.2.2.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

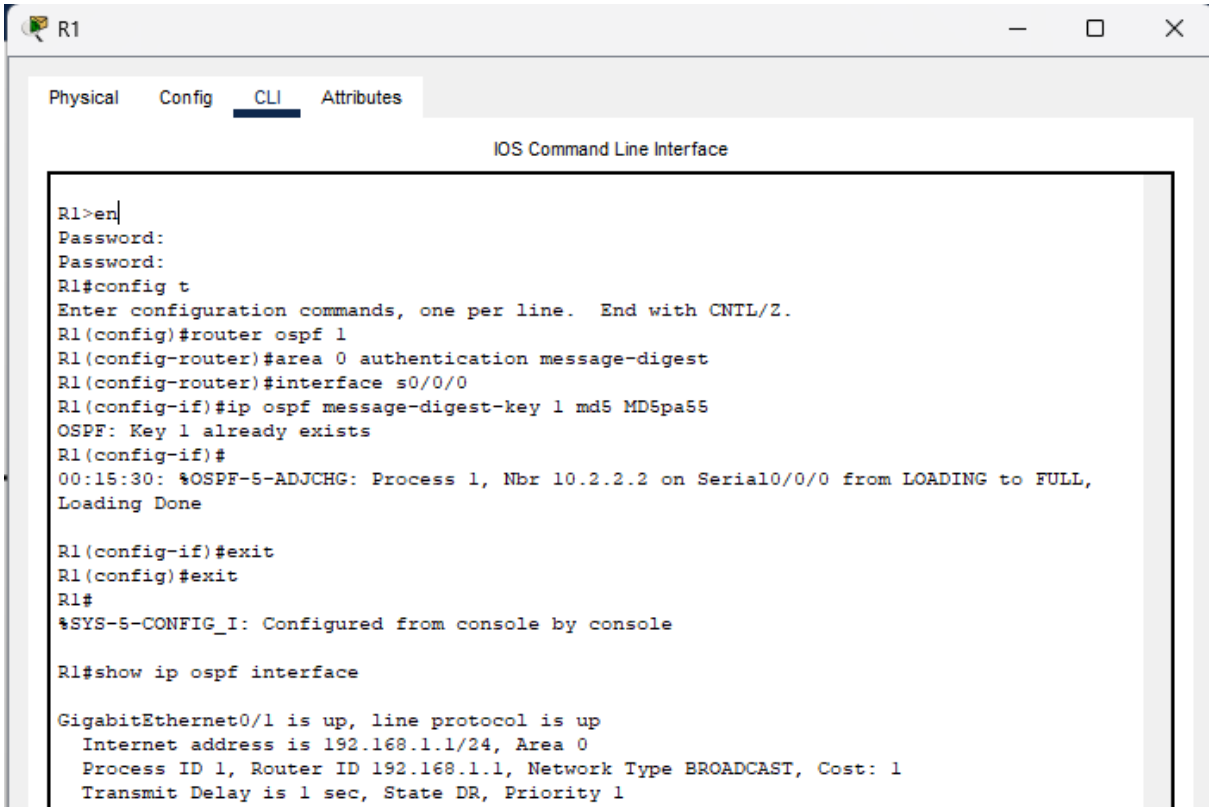
R3


R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.2.2.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10



Part 1: Configure OSPF MD5 Authentication

R1



R2

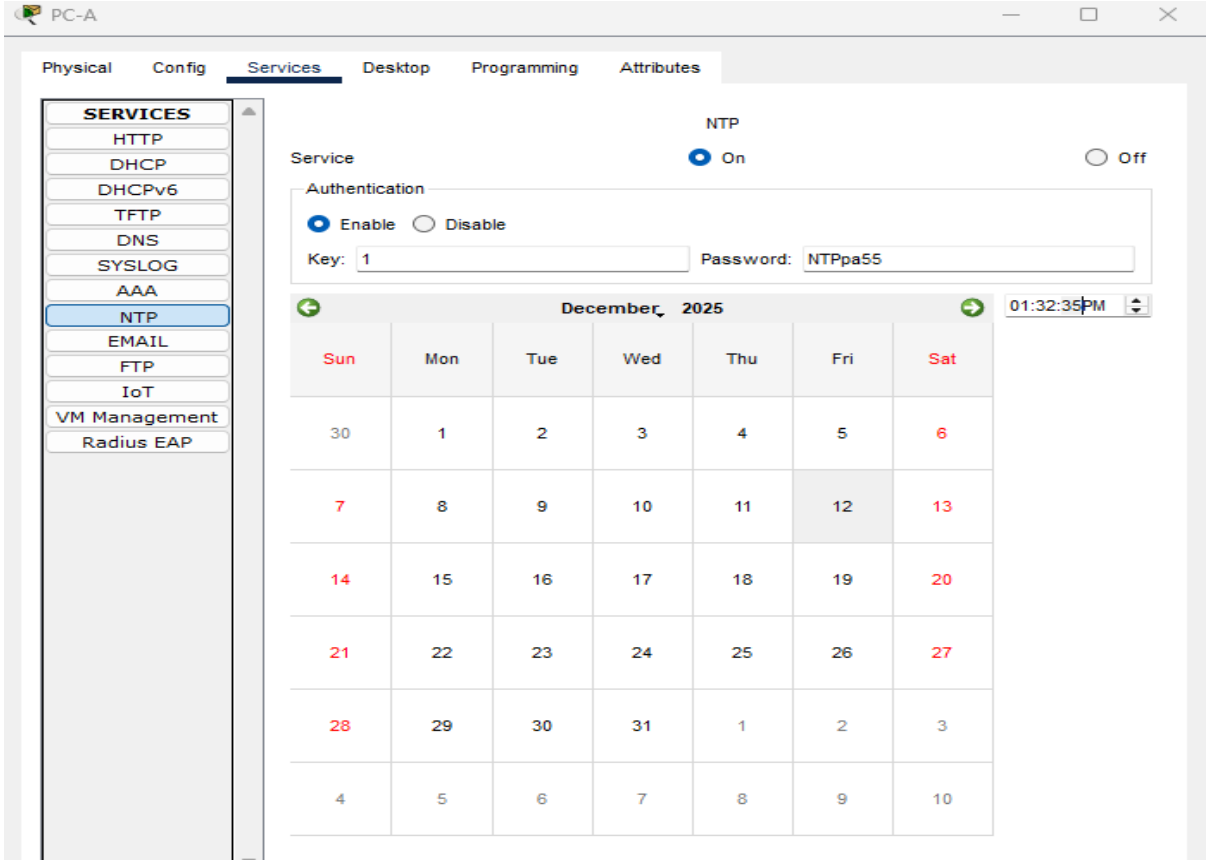
```
R2>en
Password:
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#area 0 authentication message-digest
R2(config-router)#interface s0/0/0
R2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)#interface s0/0/1
00:15:30: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL,
Loading Done

R2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)#
00:16:18: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from FULL to DOWN,
Neighbor Down: Dead timer expired

00:16:18: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from FULL to DOWN,
Neighbor Down: Interface down or detached

00:16:58: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.3.1 on Serial0/0/1 from LOADING to FULL,
Loading Done
```

R3

```
R3>en
Password:
R3#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#area 0 authentication message-digest
R3(config-router)#
00:16:10: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from FULL to DOWN,
Neighbor Down: Dead timer expired

00:16:10: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from FULL to DOWN,
Neighbor Down: Interface down or detached

R3(config-router)#interface s0/0/1
R3(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R3(config-if)#
00:16:58: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/1 from LOADING to FULL,
Loading Done
```

```
R1(config-if)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ospf interface

GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
      No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.1.1.1/30, Area 0
```

Part 2: Configure NTP



```
R1>en
Password:
R1#show clock
*0:28:47.328 UTC Mon Mar 1 1993
R1#configt
Translating "configt"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ntp server 192.168.1.5
R1(config)#ntp update-calendar
R1(config)#ntp authenticate
R1(config)#ntp trusted-key 1
R1(config)#ntp authentication-key 1 md5 NTPpa55
R1(config)#service timestamps log datetime msec
R1(config)#exit
R1#
*Dec 12, 13:36:41.3636: SYS-5-CONFIG_I: Configured from console by console
R1#show clock
13:36:56.225 UTC Fri Dec 12 2025
R1#
R1#
```

Part 3: Configure Routers to Log Messages to the Syslog Server

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
*Dec 12, 13:36:41.3636: SYS-5-CONFIG_I: Configured from console by console
R1#show clock
13:36:56.225 UTC Fri Dec 12 2025
R1#
R1#
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging host 192.168.1.6
R1(config)#exit
R1#
*Dec 12, 13:39:53.3939: SYS-5-CONFIG_I: Configured from console by console
*Dec 12, 13:39:53.3939:  %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.6 port
514 started - CLI initiated
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
          0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.


No Inactive Message Discriminator.
```

PC-B

Physical Config Services Desktop Programming Attributes

SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP

Syslog

Syslog

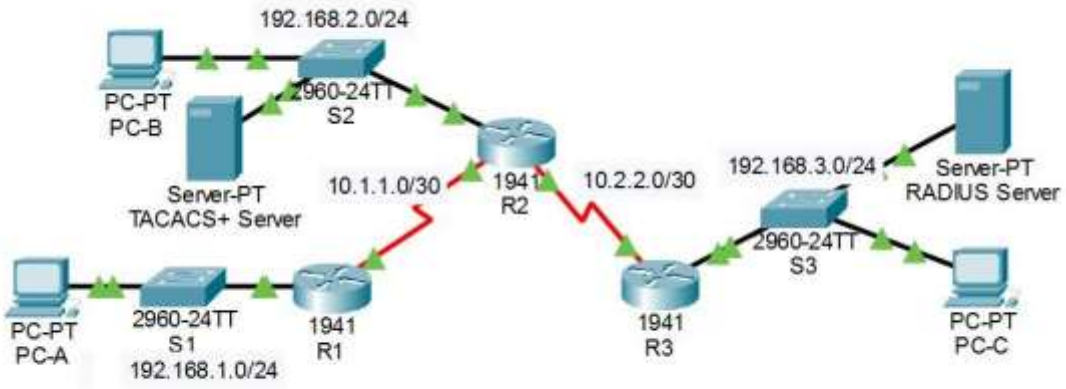
Service On Off

	Time	HostName	Message
1	12.12.2025 01:39:53.602 PM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console by...
2	12.12.2025 01:39:53.602 PM	192.168.1.1	: %SYS-6-LOGGINGHOST_STARTSTO...

Practical - 2

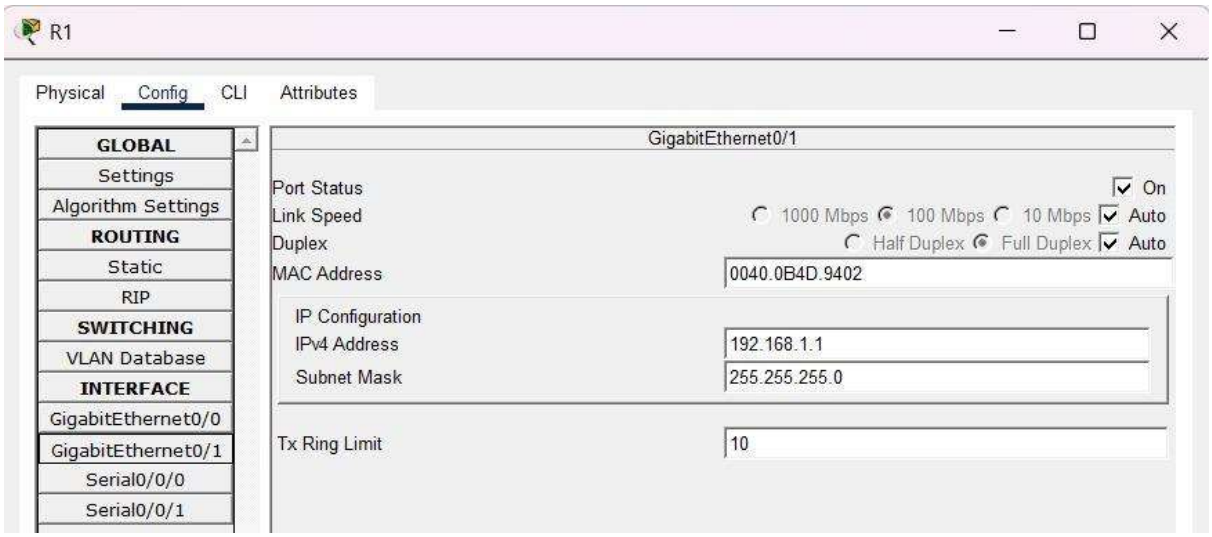
Configure AAA Authentication

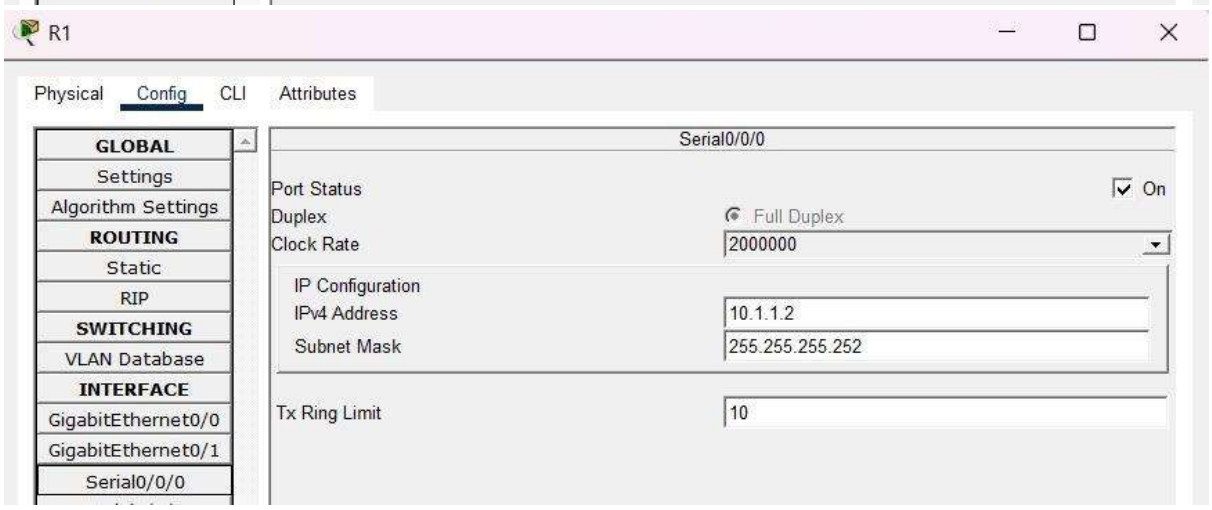
Topology



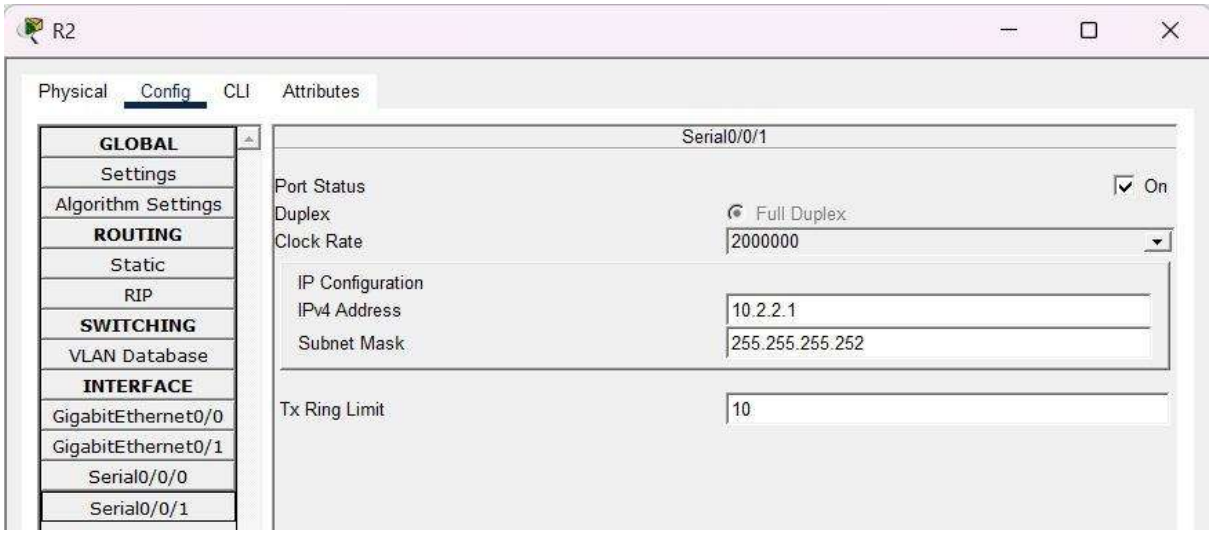
Configurations:

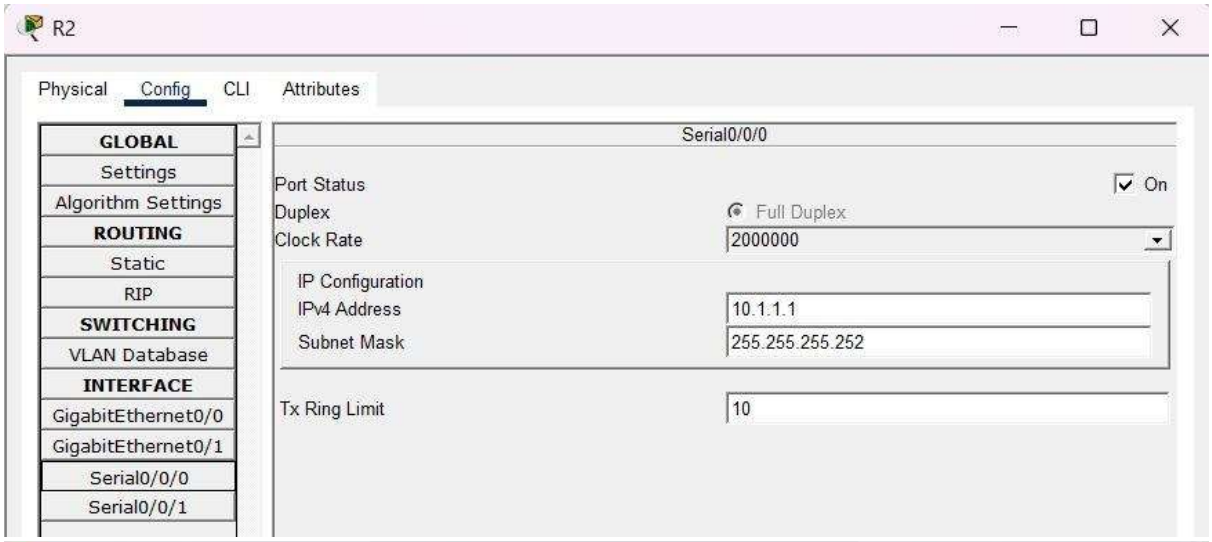
R1

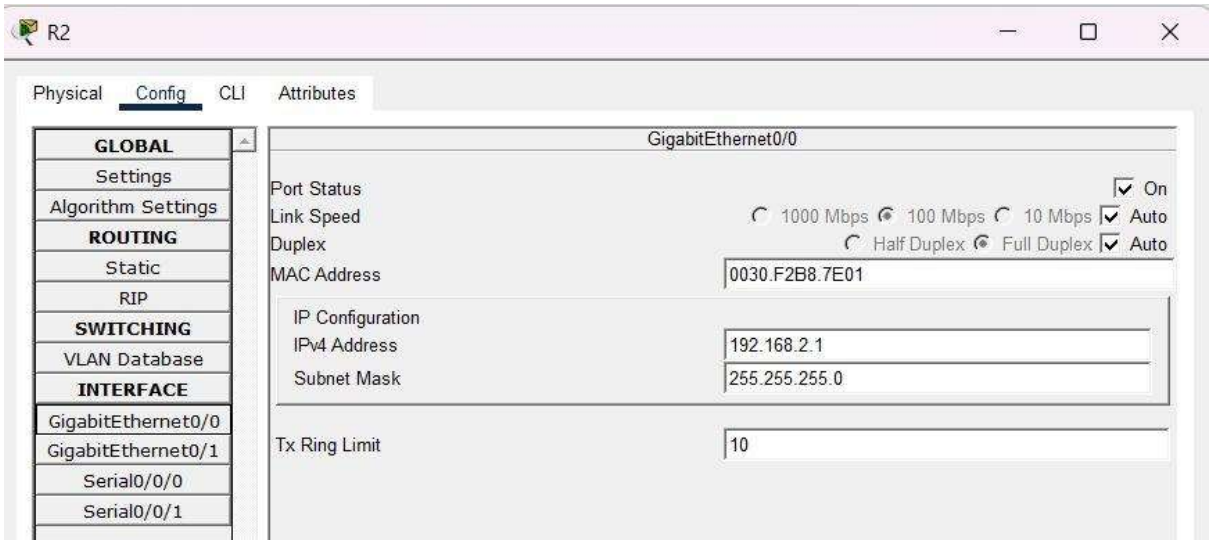




R2

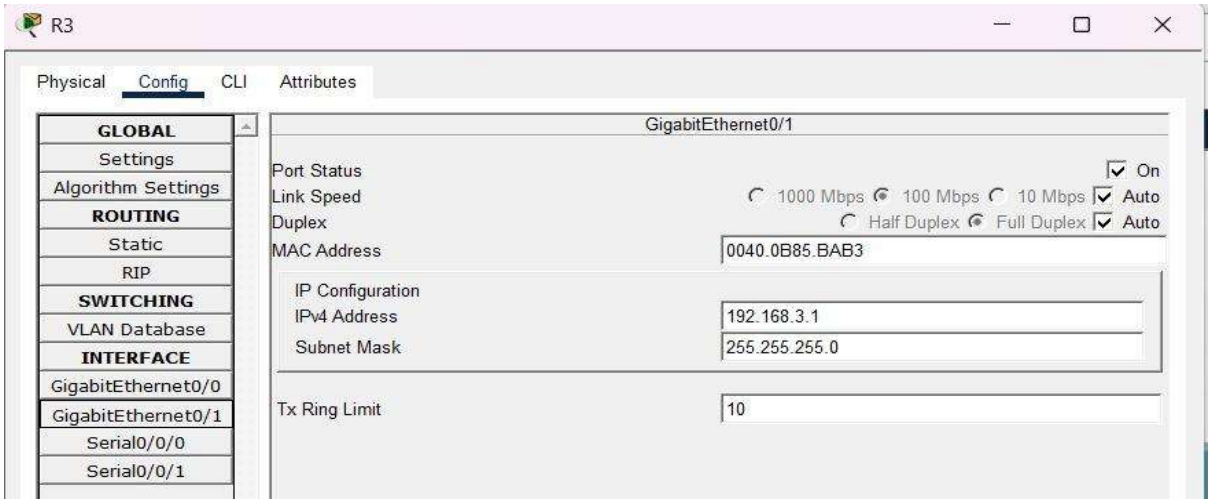

R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.2.2.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

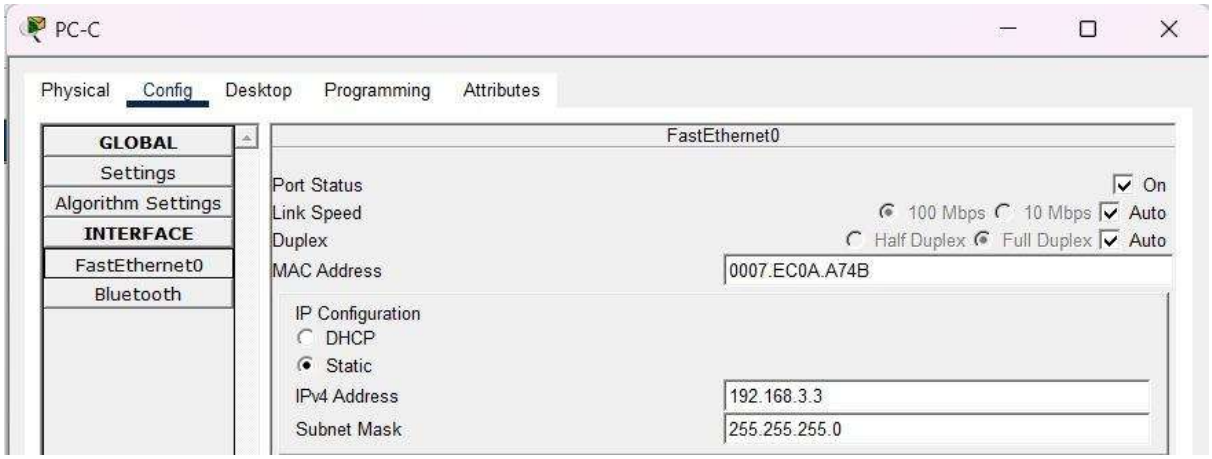

R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/0
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.F2B8.7E01
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

R3

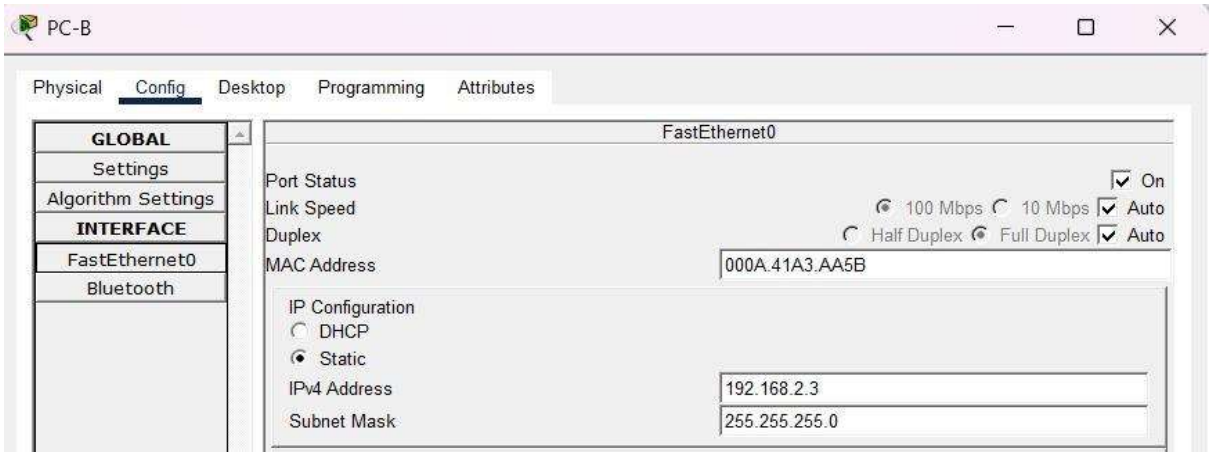
R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.2.2.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10


R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/1
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0040.0B85.BAB3
IP Configuration
IPv4 Address
192.168.3.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

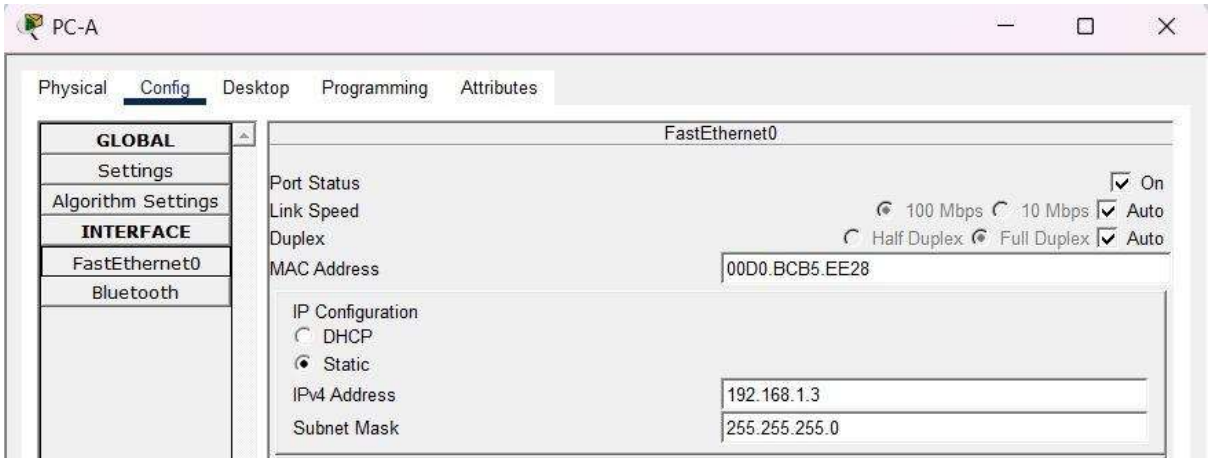
PC-C


PC-C
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0007.EC0A.A74B
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.3
Subnet Mask
255.255.255.0

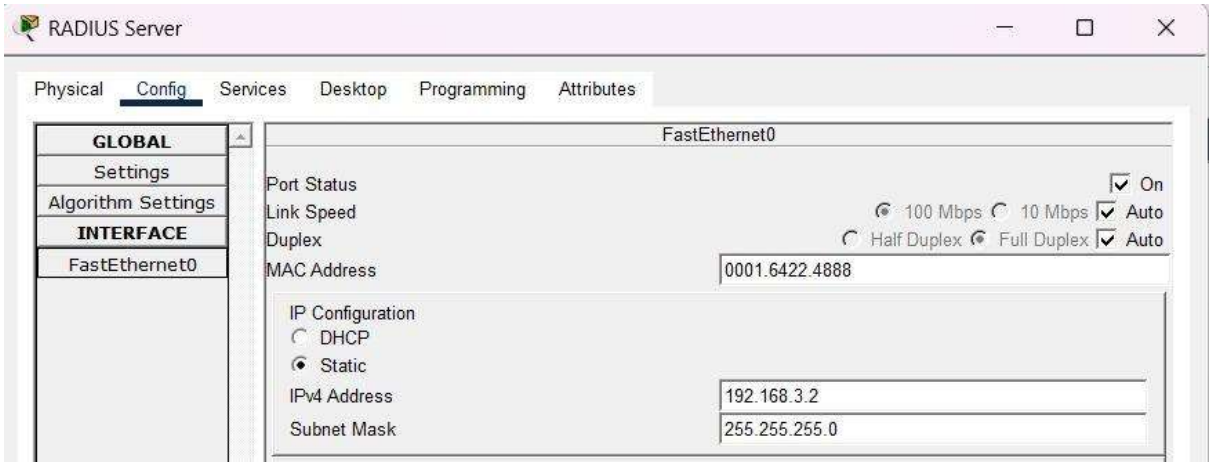
PC-B



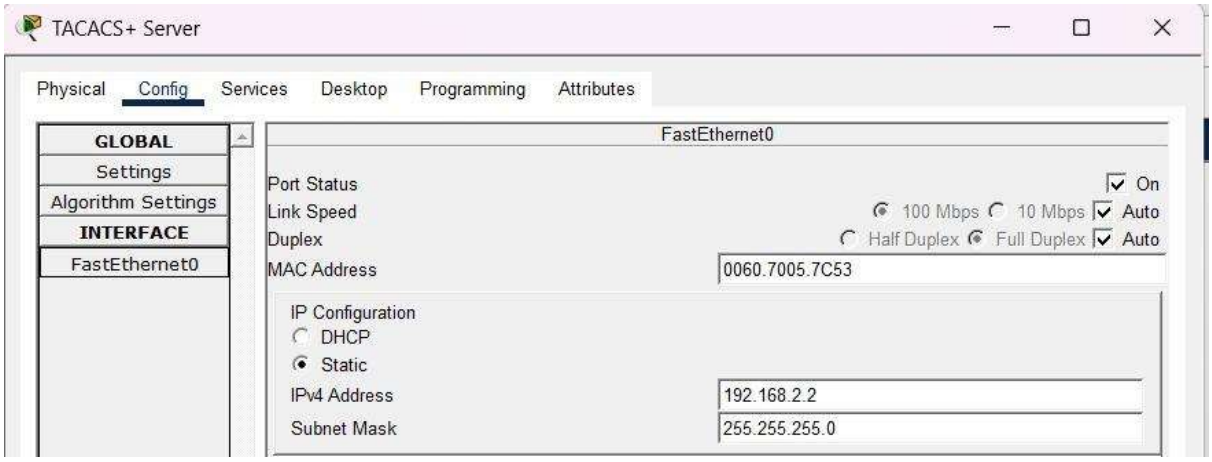
PC-A



Server 1

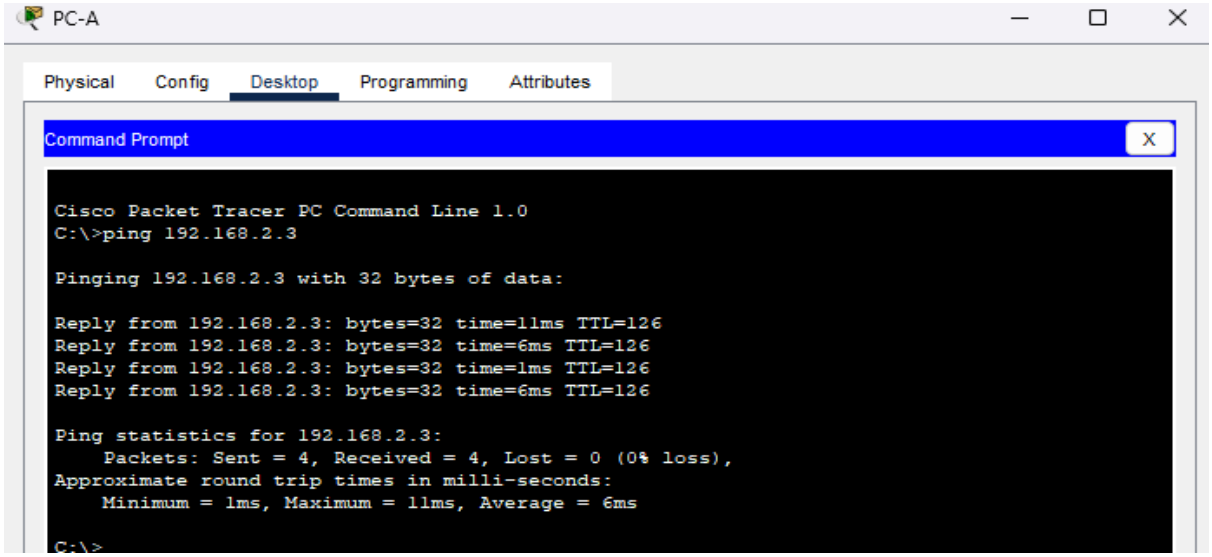


Server 2

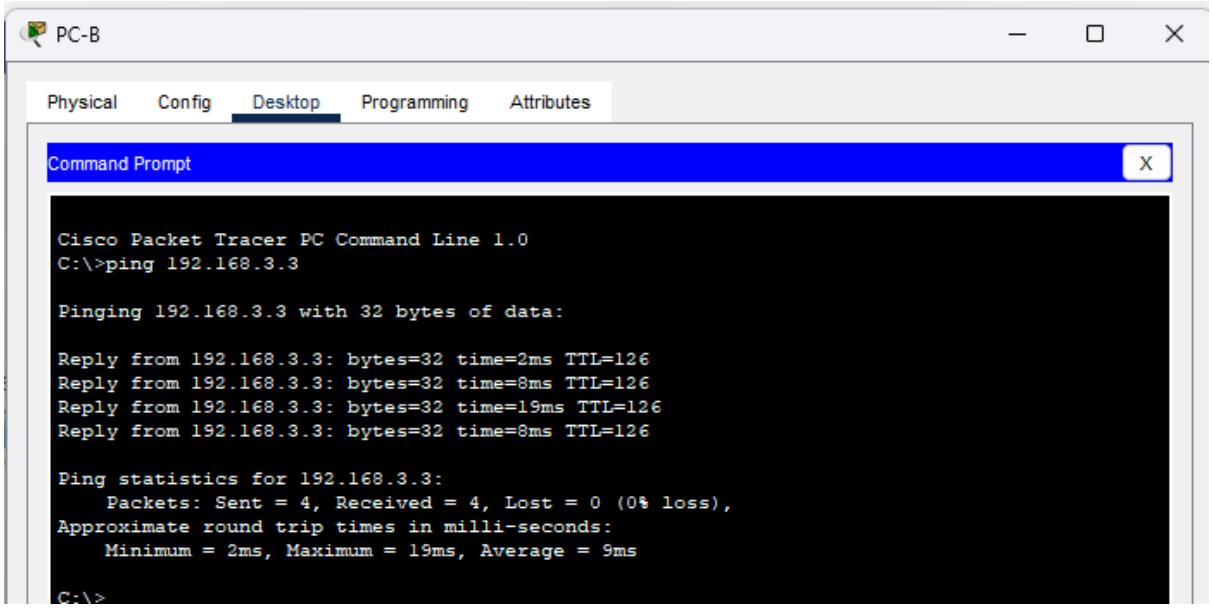


Part 1: Configure Local AAA Authentication for Console Access on R1

PC-A -> PC-B



PC-B -> PC-C



PC-C -> PC-A

PC-C

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=10ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=13ms TTL=125
Reply from 192.168.1.3: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 13ms, Average = 8ms

C:\>
```

R1

```
R1>en
Password:
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#username Admin1 secret admin1pa55
R1(config)#aaa new-model
R1(config)#aaa authentication login default local
R1(config)#line console 0
R1(config-line)#login authentication default
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#exit
```

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
************ AUTHORIZED ACCESS ONLY *************
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.


************ AUTHORIZED ACCESS ONLY *************
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.


User Access Verification

Username: Admin1
Password:
R1>en
```

Part 2: Configure Local AAA Authentication for vty Lines on R1

```
R1>en
Password:
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip domain-name ccnasecurity.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#aaa authentication login SSH-LOGIN local
*Mar 1 0:27:49.694: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

PC-A

PC-A

Physical Config Desktop Programming Attributes

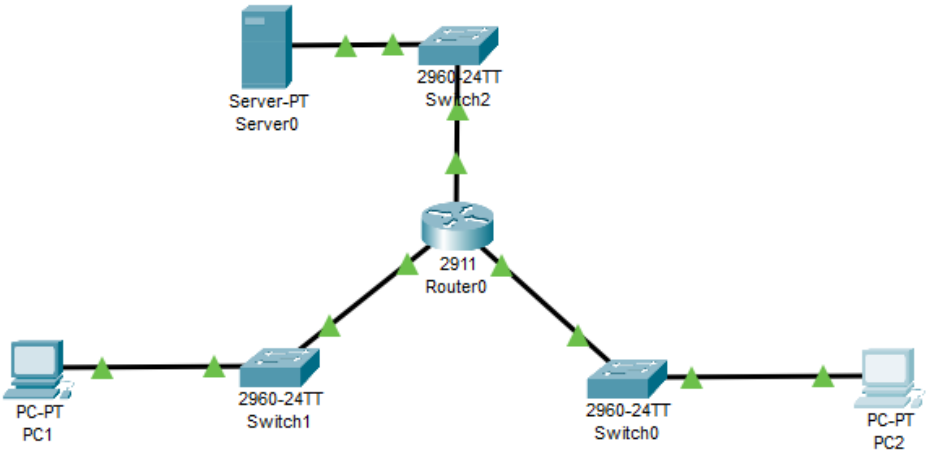
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=11ms TTL=126
Reply from 192.168.2.3: bytes=32 time=6ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>ssh -l Admin1 192.168.1.1

Password:
R1>
```

Practical - 3

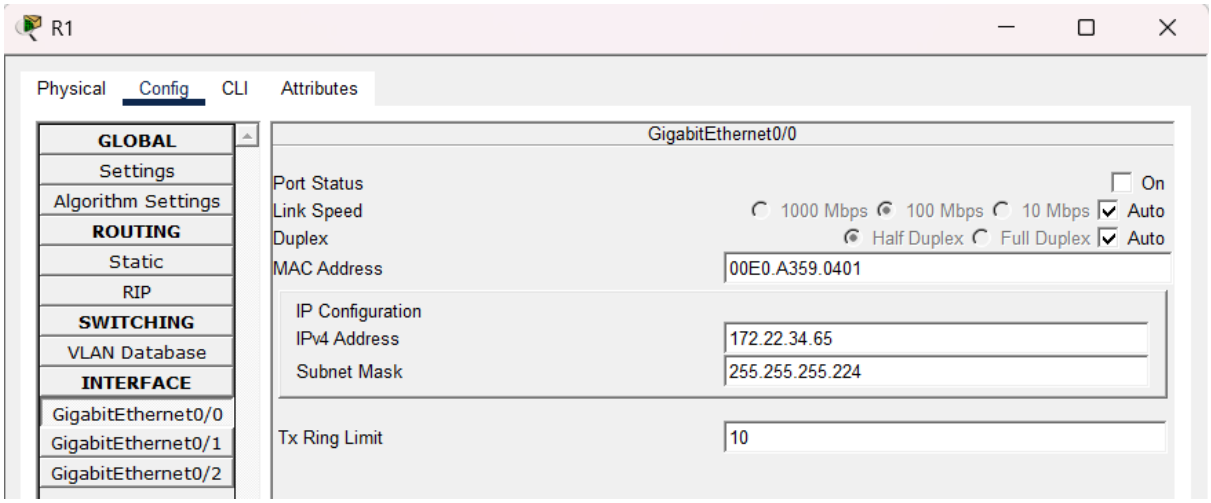
Configuring Extended ACLs

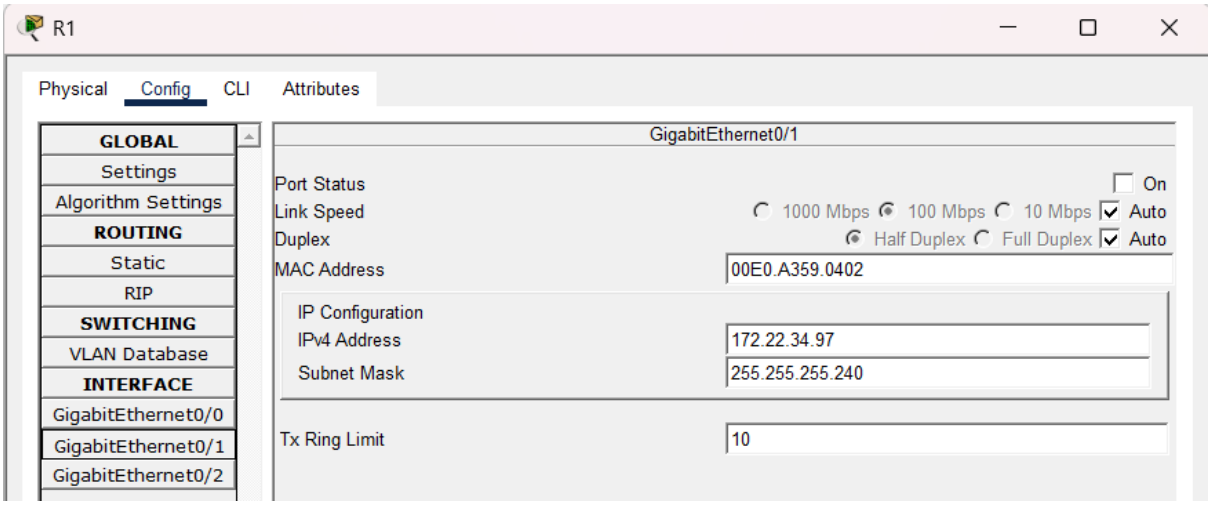
Topology

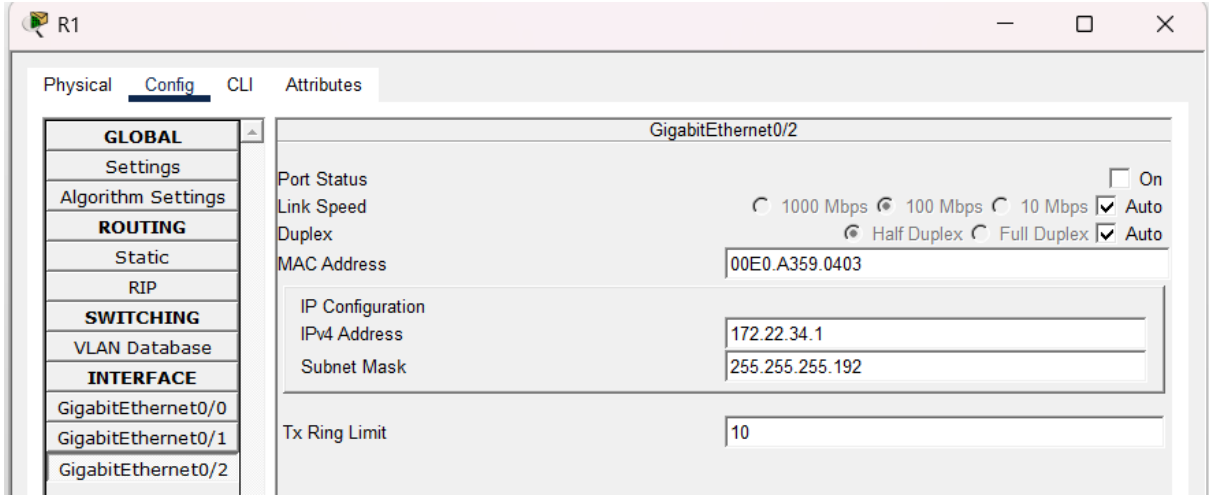


Configurations:

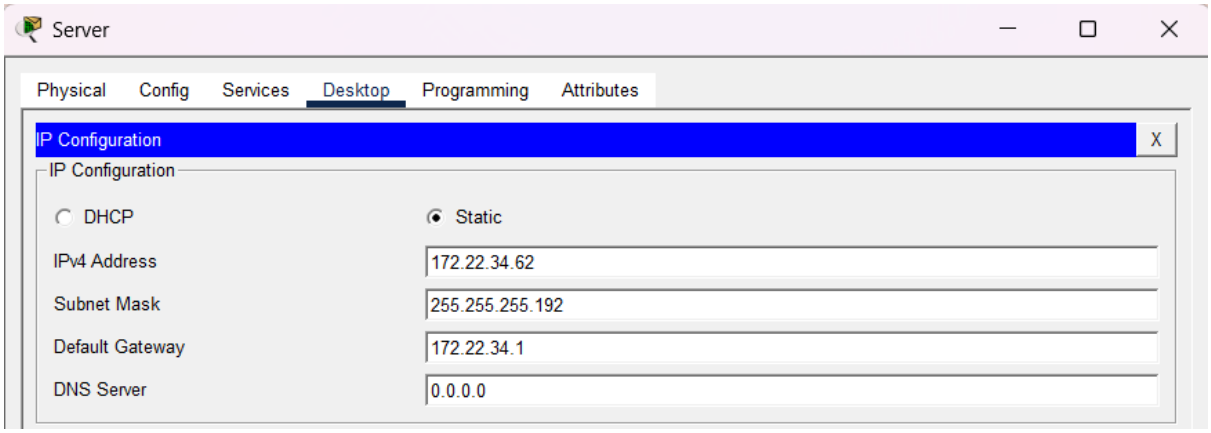
R1



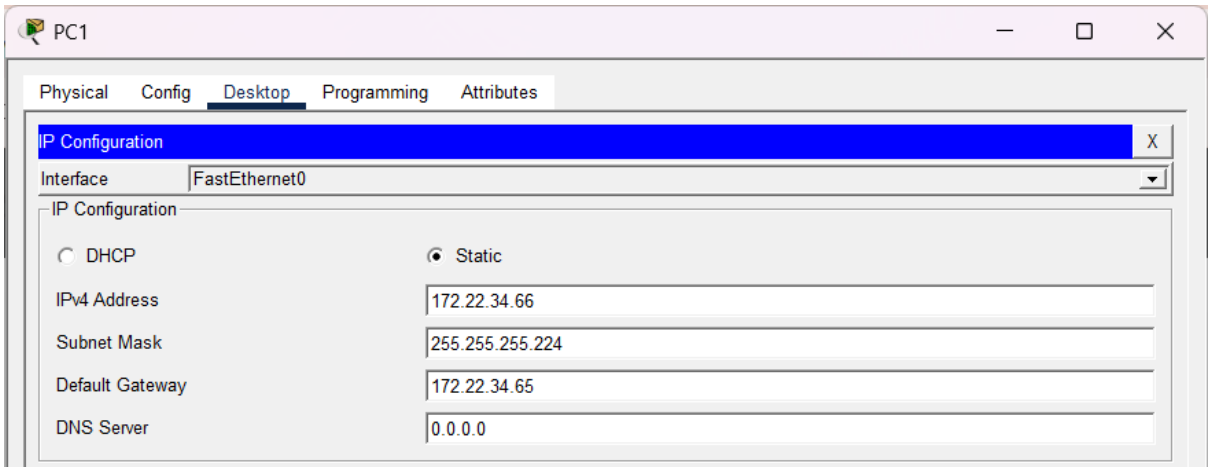



R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/2
GigabitEthernet0/2
Port Status
On
Link Speed
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.A359.0403
IP Configuration
IPv4 Address
172.22.34.1
Subnet Mask
255.255.255.192
Tx Ring Limit
10

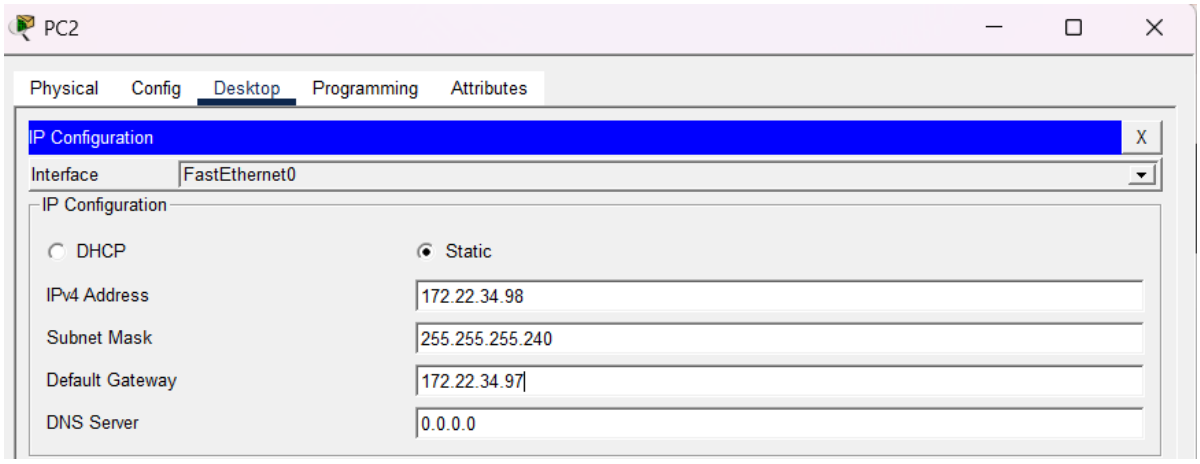
Server


Server
Physical
Config
Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
172.22.34.62
Subnet Mask
255.255.255.192
Default Gateway
172.22.34.1
DNS Server
0.0.0.0

PC1

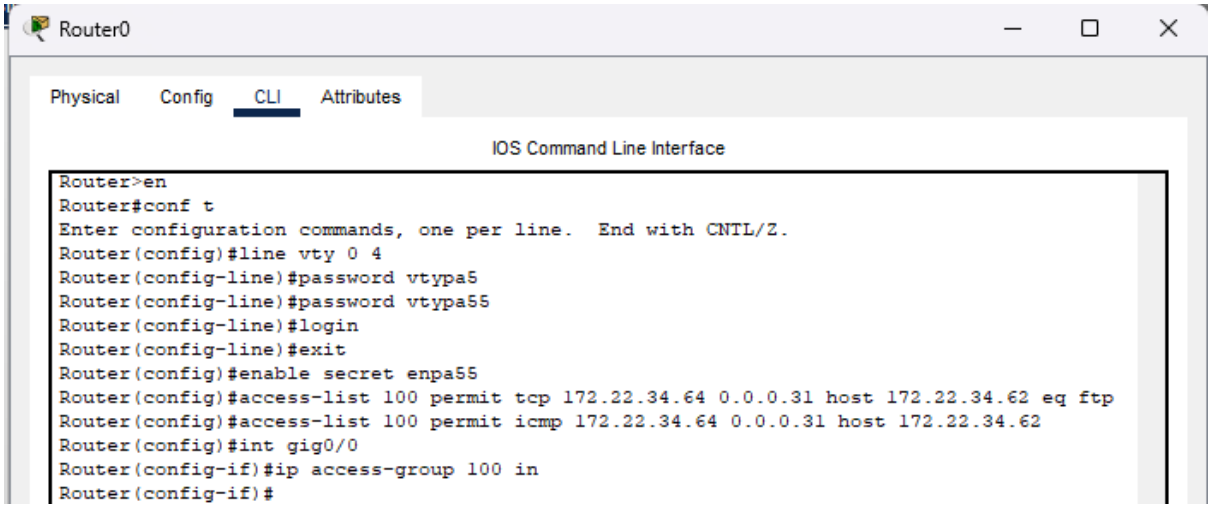

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
172.22.34.66
Subnet Mask
255.255.255.224
Default Gateway
172.22.34.65
DNS Server
0.0.0.0

PC2

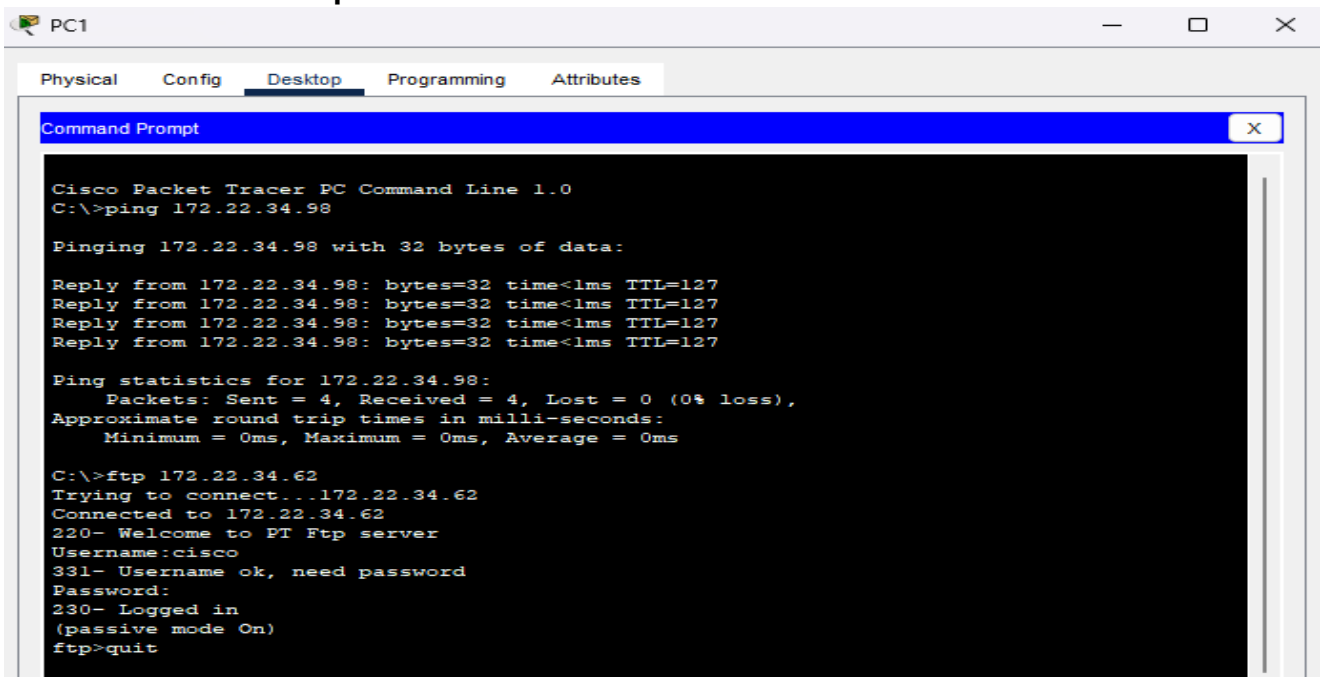


Part 1: Configure, Apply and Verify an Extended Numbered ACL

R0



PC-1 Command Prompt



PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
221- Service closing control connection.
C:\>ping 172.22.34.98

Pinging 172.22.34.98 with 32 bytes of data:

Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.

Ping statistics for 172.22.34.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.22.34.62

Pinging 172.22.34.62 with 32 bytes of data:

Reply from 172.22.34.62: bytes=32 time<1ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.22.34.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Part 2: Configure, Apply and Verify an Extended Named ACL

R0

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Password:
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended HTTP_ONLY
Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.64 eq www
Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
Router(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62
Router(config-ext-nacl)#exit
Router(config)#int gig0/1
Router(config-if)#ip access-group HTTP_ONLY in
Router(config-if)#exit
```

PC1 Browser

PC1

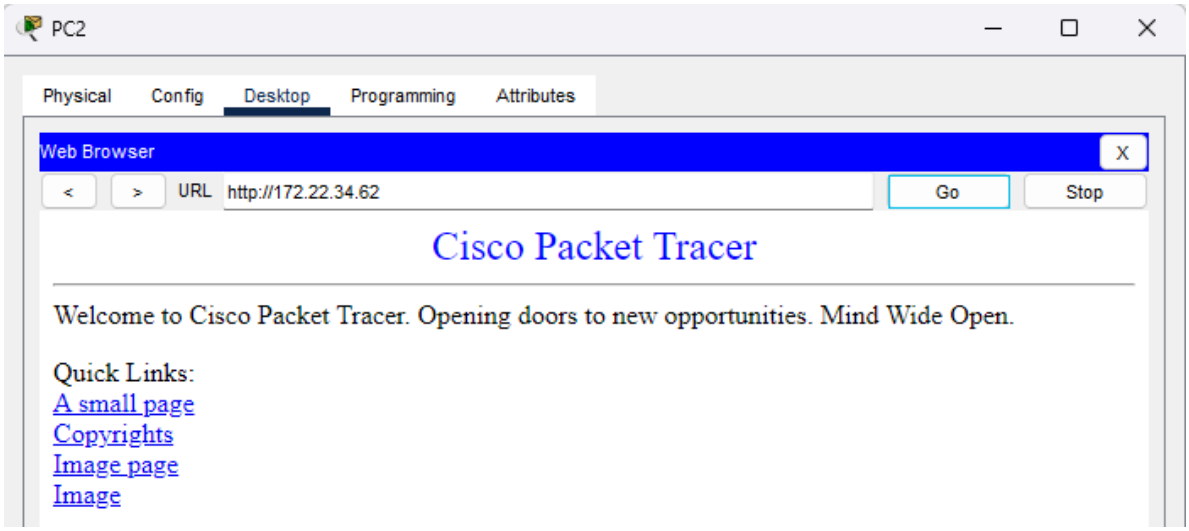
Physical Config Desktop Programming Attributes

Web Browser

< > URL http://172.22.34.62 Go Stop

Request Timeout

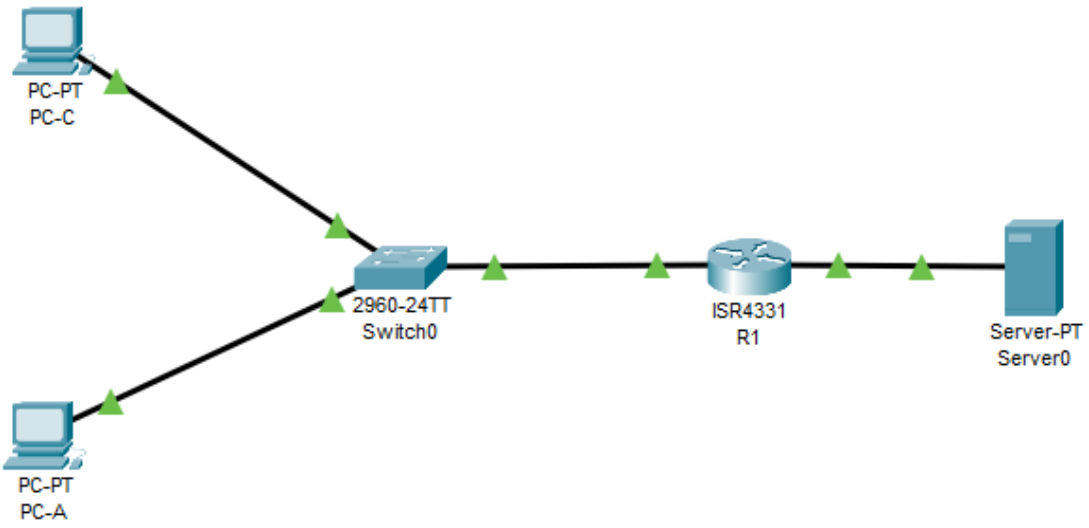
PC2 Browser


PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://172.22.34.62
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

Practical - 4

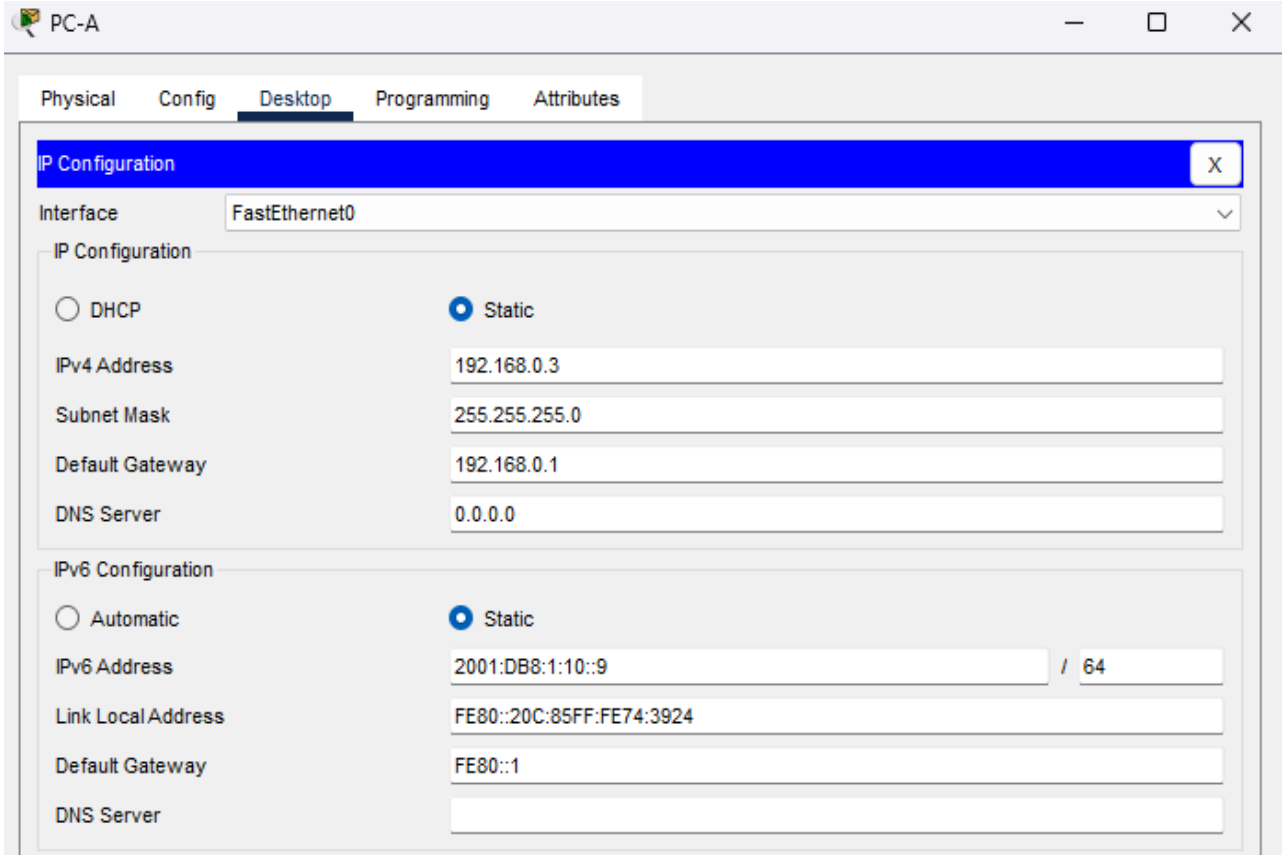
Configure IP ACLs to Mitigate Attacks and IPV6 ACLs

Topology

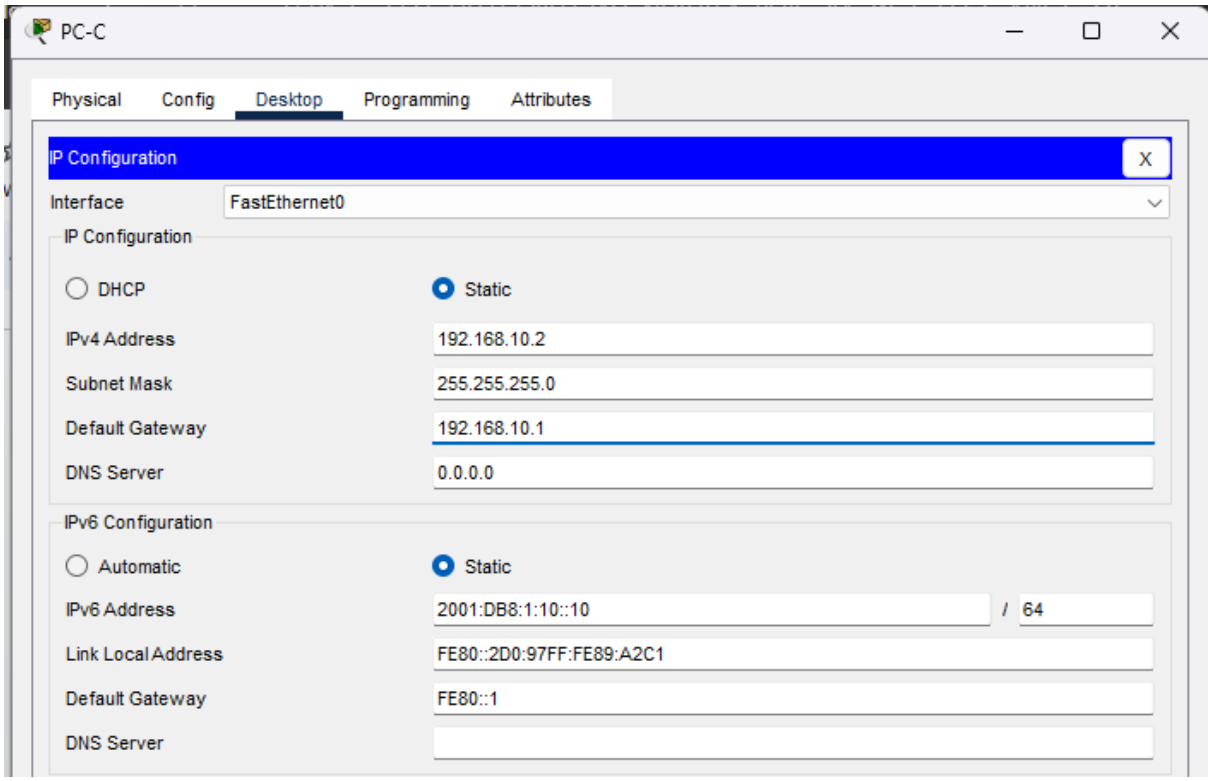


Configurations:

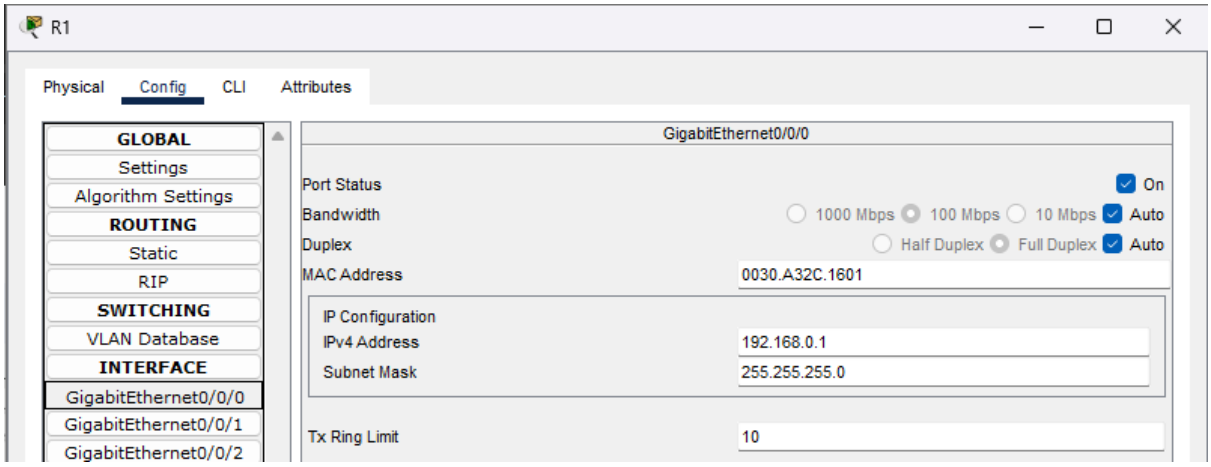
PC-A

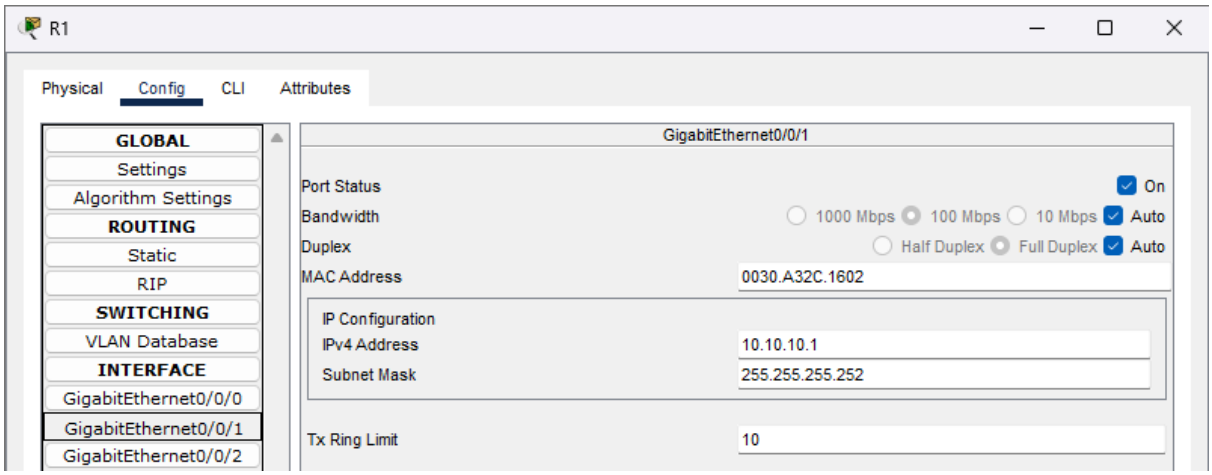


PC-C

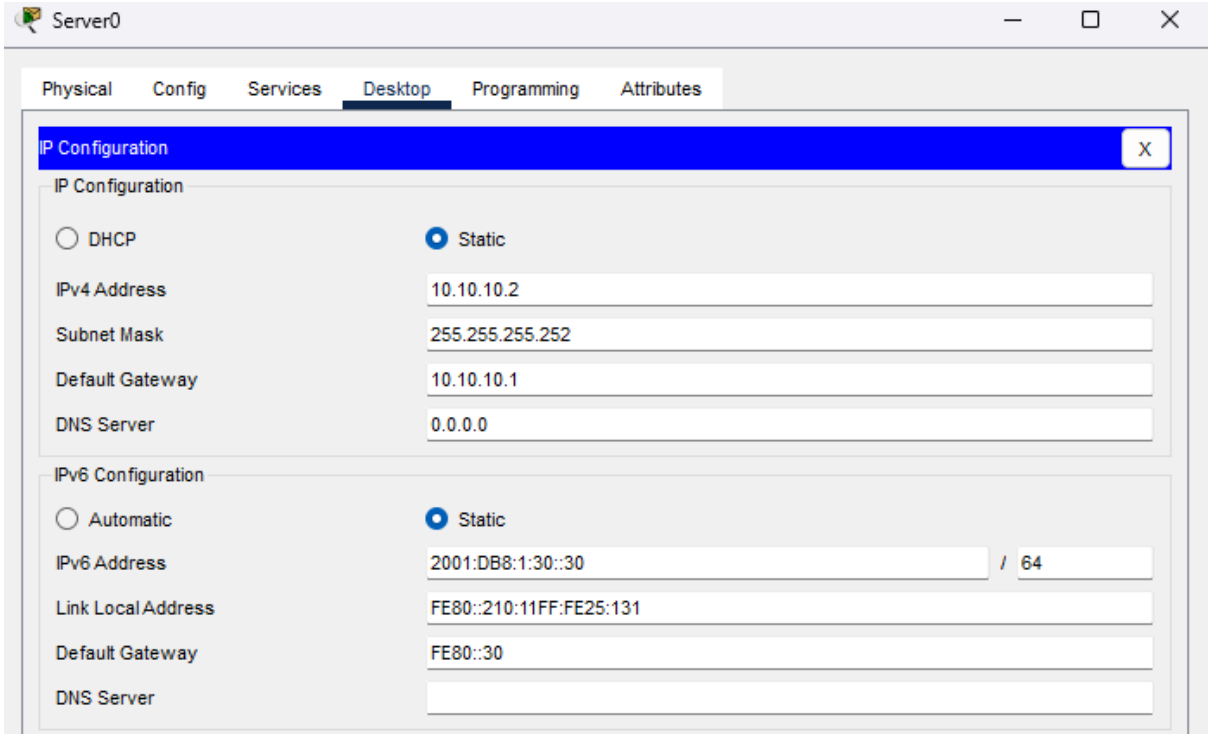

PC-C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1:10::10
/ 64
Link Local Address
FE80::2D0:97FF:FE89:A2C1
Default Gateway
FE80::1
DNS Server

R1

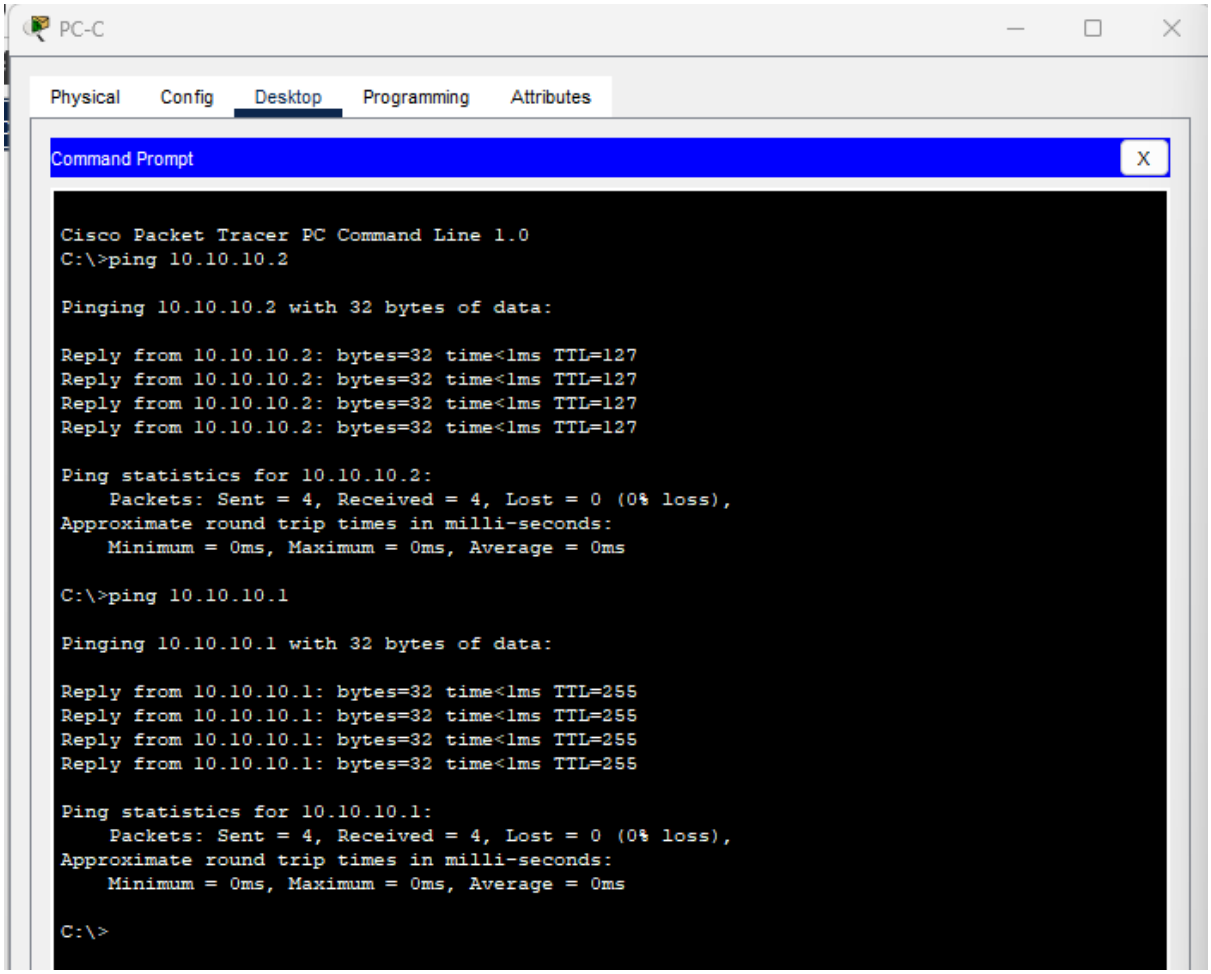

R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
GigabitEthernet0/0/2
GigabitEthernet0/0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.A32C.1601
IP Configuration
IPv4 Address
192.168.0.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10


R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
GigabitEthernet0/0/2
GigabitEthernet0/0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0030.A32C.1602
IP Configuration
IPv4 Address
10.10.10.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

Server 0



PC-C Command Prompt



PC-A Command Prompt

PC-A

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127
Reply from 10.10.10.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:

Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255
Reply from 10.10.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

R1

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret ciscoenpa55
Router(config)#line console 0
Router(config-line)#password conpa55
Router(config-line)#login
Router(config-line)#exit
Router(config)#ip domain-name ccnasecurity.com
Router(config)#username admin secret adminpa55
Router(config)#line vty 0 4
Router(config-line)#login local
Router(config-line)#exit
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
```

```
Router(config)#hostname R1
R1(config)#
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

```
R1(config)#access-list 1 deny any
R1(config)#
```

PC-C Command Prompt

```
C:\>ssh -l admin 192.168.0.1

Password:



R1>
```

PC-A Command Prompt

```
C:\>ssh -l admin 192.168.0.1

Password:



R1>
```

R1 CLI

```
R1(config)#line vty 0 4
R1(config-line)#access-class 1 in
R1(config-line)#
```

PC-C Command Prompt

```
[Connection to 192.168.0.1 closed by foreign host]
C:\>ssh -l admin 192.168.0.1

Password:



R1>
```

PC-A Command Prompt

```
[Connection to 192.168.0.1 closed by foreign host]
C:\>ssh -l admin 192.168.0.1

% Connection refused by remote host
C:\>
```

R1 CLI

```
R1(config-line)#access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq ftp
R1(config)#access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq ftp
R1(config)#access-list 120 permit tcp host 192.168.0.3 host 10.10.10.2 eq www
R1(config)#access-list 120 permit tcp host 192.168.0.2 host 10.10.10.2 eq 22
R1(config)#int gi 0/0/0
R1(config-if)#ip access-group 120 in
R1(config-if)#
```

PC-A Command Prompt

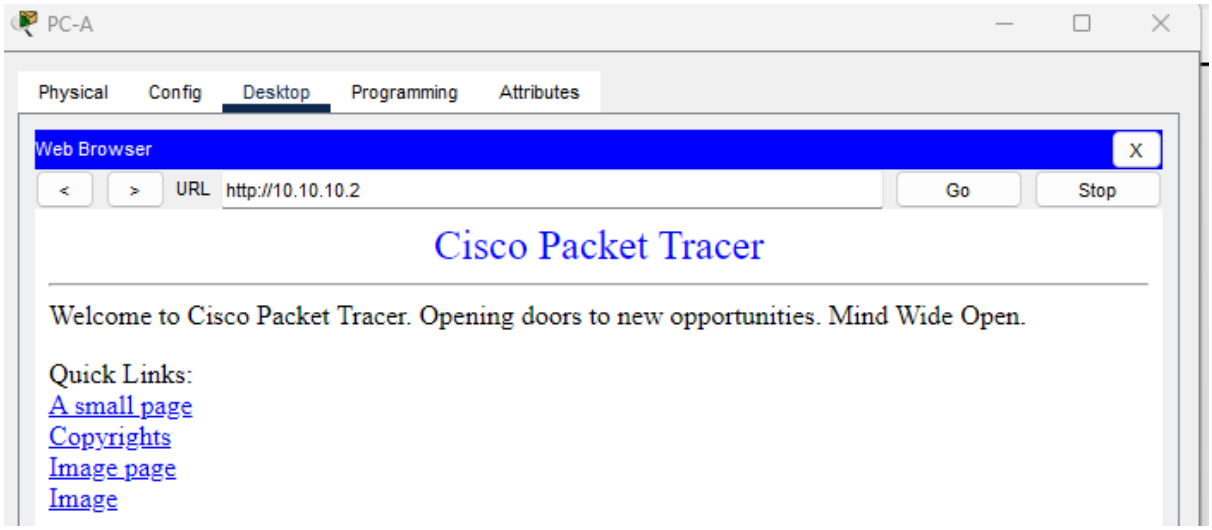
```
C:\>ftp 10.10.10.2
Trying to connect...10.10.10.2
Connected to 10.10.10.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

[Password: cisco]

PC-C Command Prompt

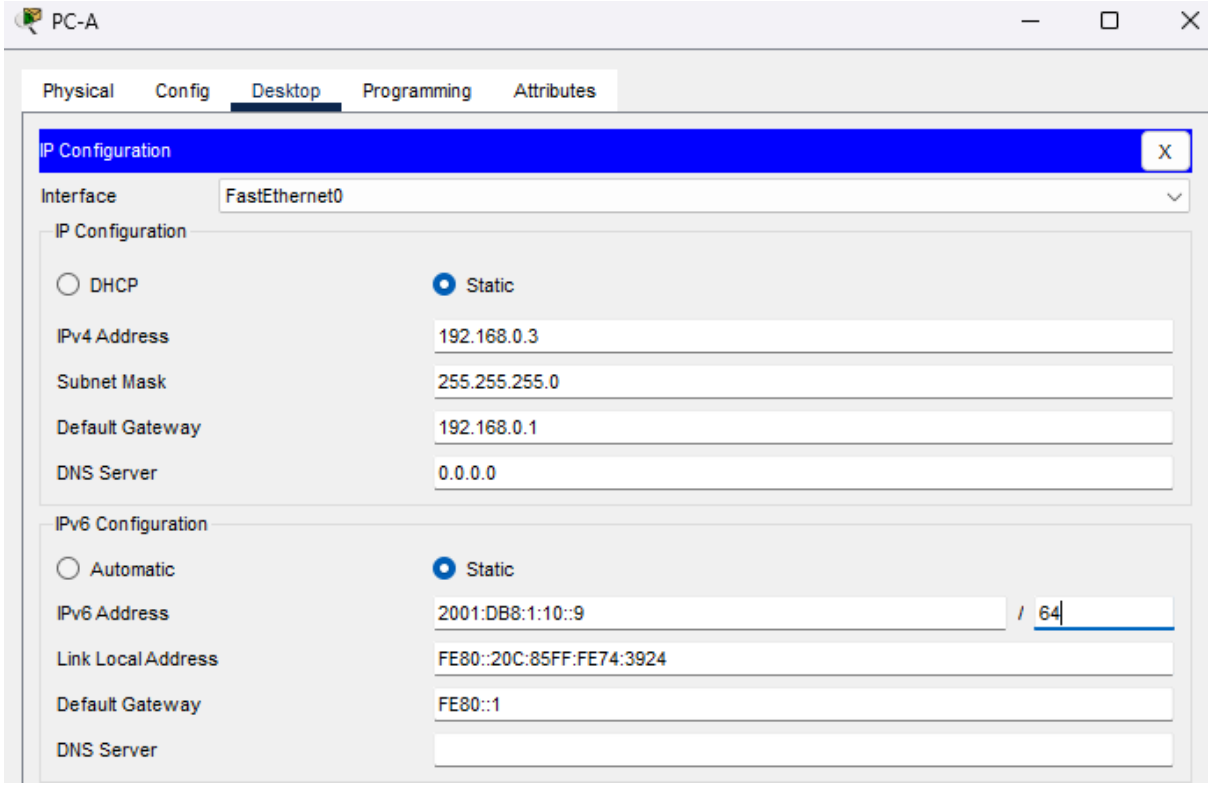
```
C:\>ftp 10.10.10.2
Trying to connect...10.10.10.2

%Error opening ftp://10.10.10.2/ (Timed out)
.



(Disconnecting from ftp server)
```

PC-A Browser

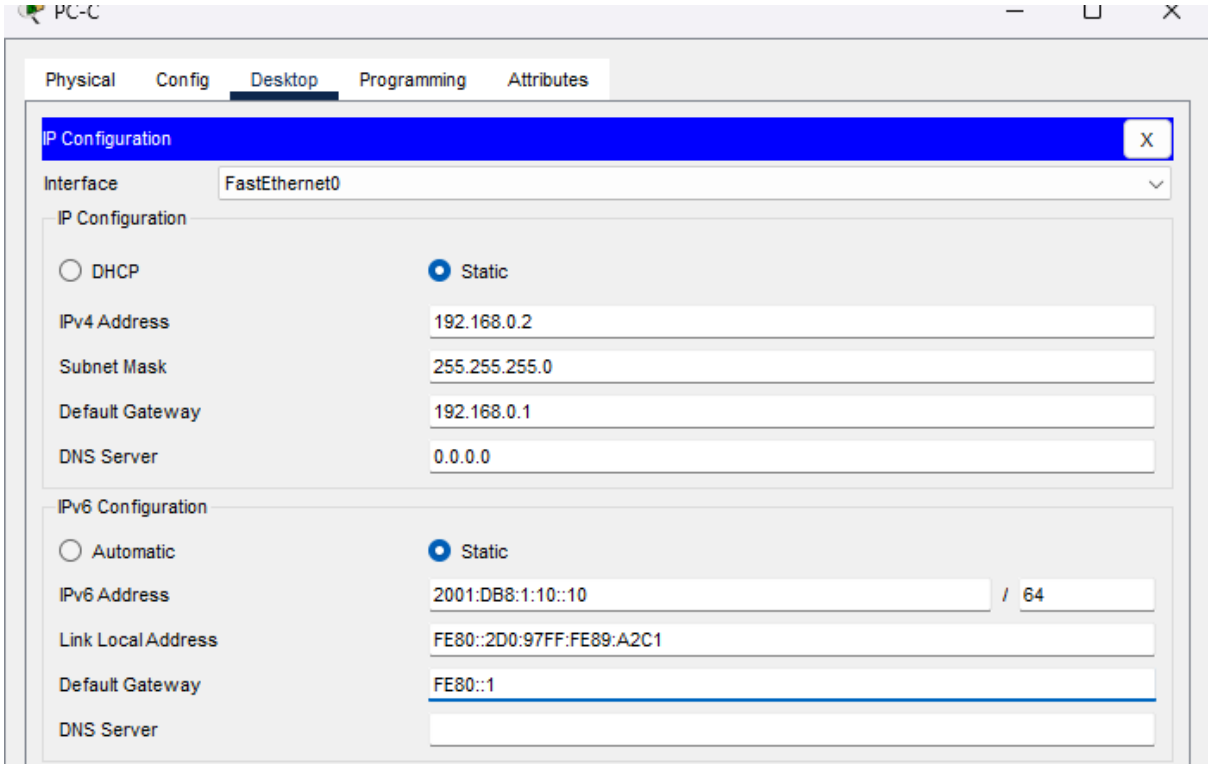


IPv6 Address

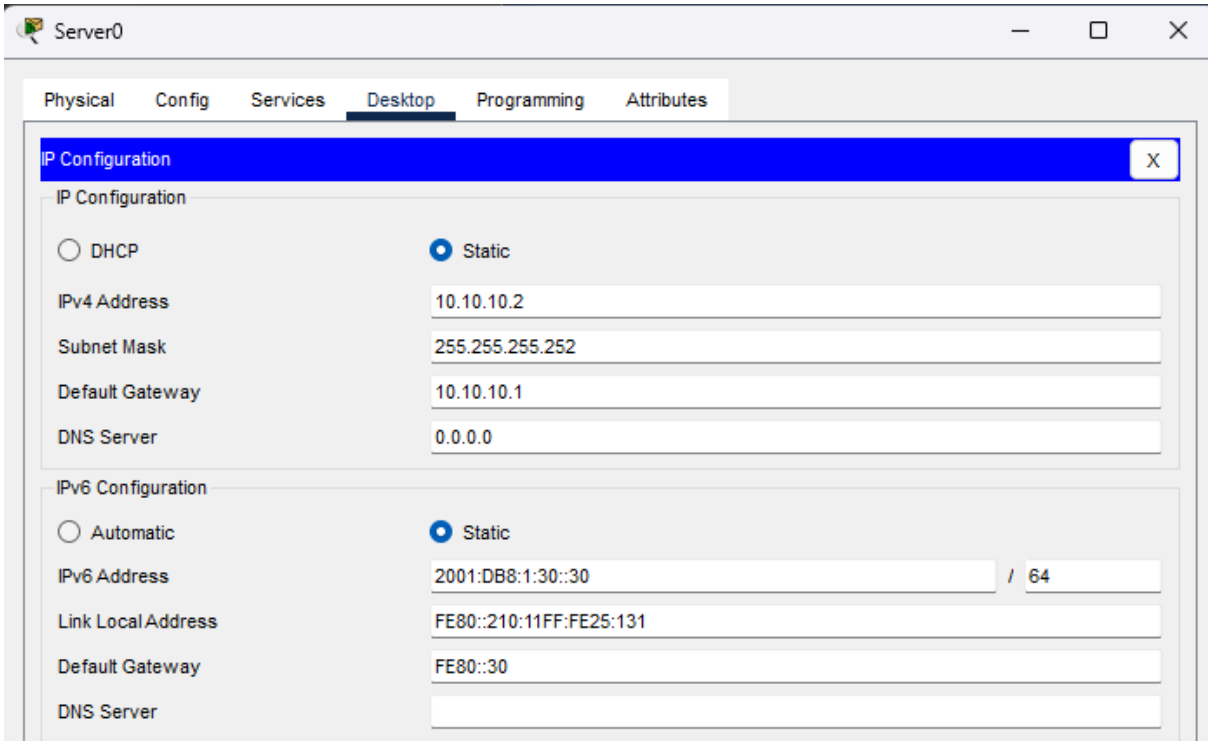
PC-A



PC-C



Server 0

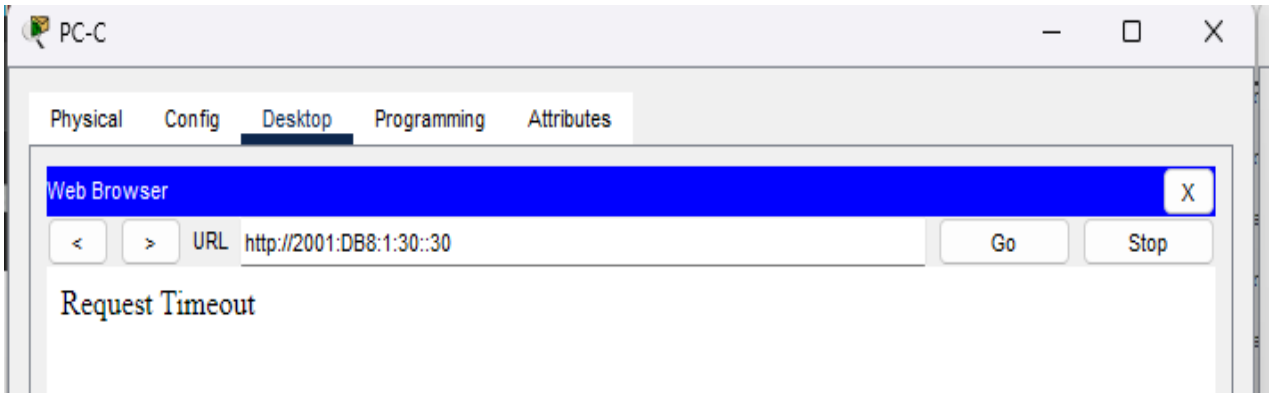


[Default Local Address, Not to change]

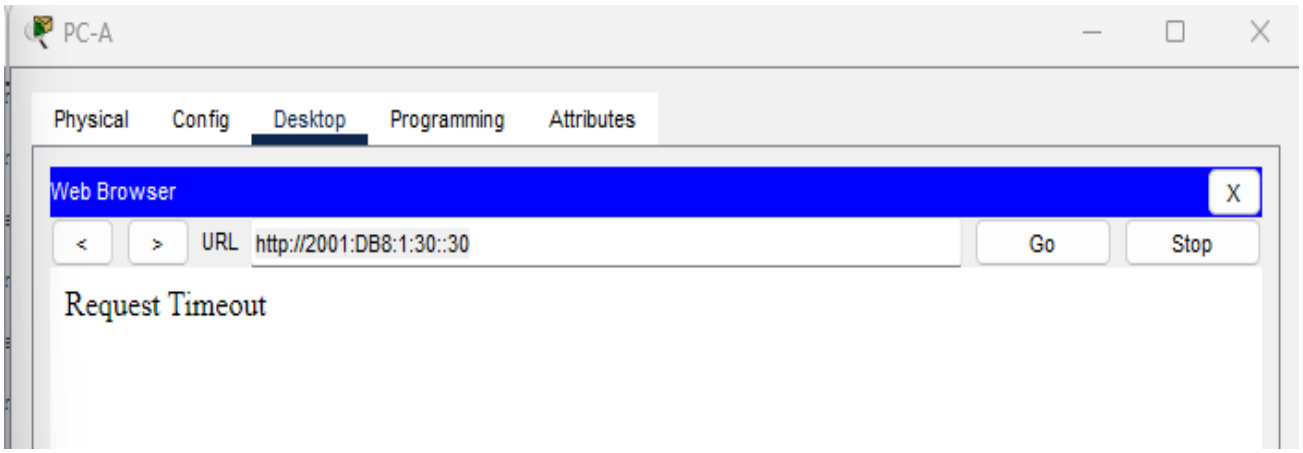
R1 CLI

```
R1(config)#access-list 120 permit tcp host 192.168.0.2 host 192.168.0.1 eq 22
R1(config)#interface gigabitethernet 0/0/0
R1(config-if)#ipv6 address 2001:DB8:1:10::1/64
R1(config-if)#ipv6 address FE80::1 link-local
R1(config-if)#ipv6 unicast-routing
R1(config)#ipv6 route 2001:DB8:1:10::1/64 2001:DB8:1:30::30
R1(config)#
R1(config)#ipv6 access-list HTTP
R1(config-ipv6-acl)#permit tcp host 2001:DB8:1:10::9 host 2001:DB8:1:30::30 eq www
R1(config-ipv6-acl)#permit tcp host 2001:DB8:1:10::9 host 2001:DB8:1:30::30 eq 443
R1(config-ipv6-acl)#exit
R1(config)#interface gigabitethernet 0/0/0
R1(config-if)#ipv6 traffic-filter HTTP in
R1(config-if)#interface GigabitEthernet0/0/0
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0/1
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0/2
R1(config-if)#exit
R1(config)#interface GigabitEthernet0/0/1
R1(config-if)#
```

PC-C Browser



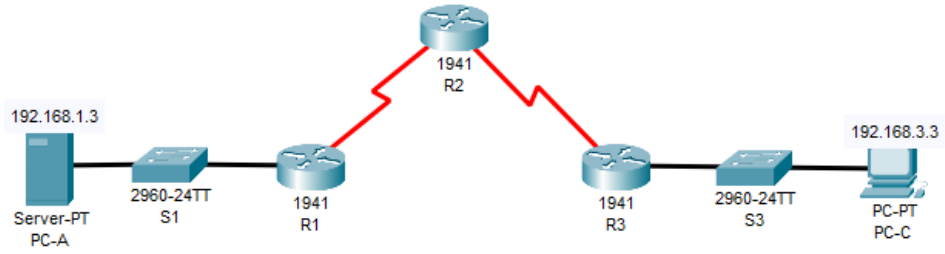
PC-A Browser



Practical - 5

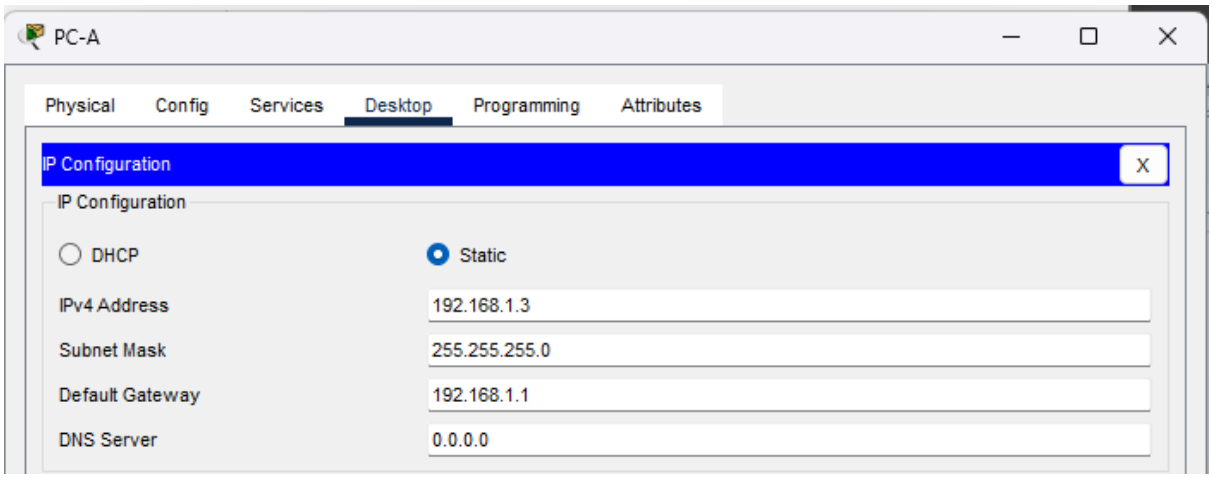
Configuring a Zone-Based Policy Firewall (ZPF)

Topology

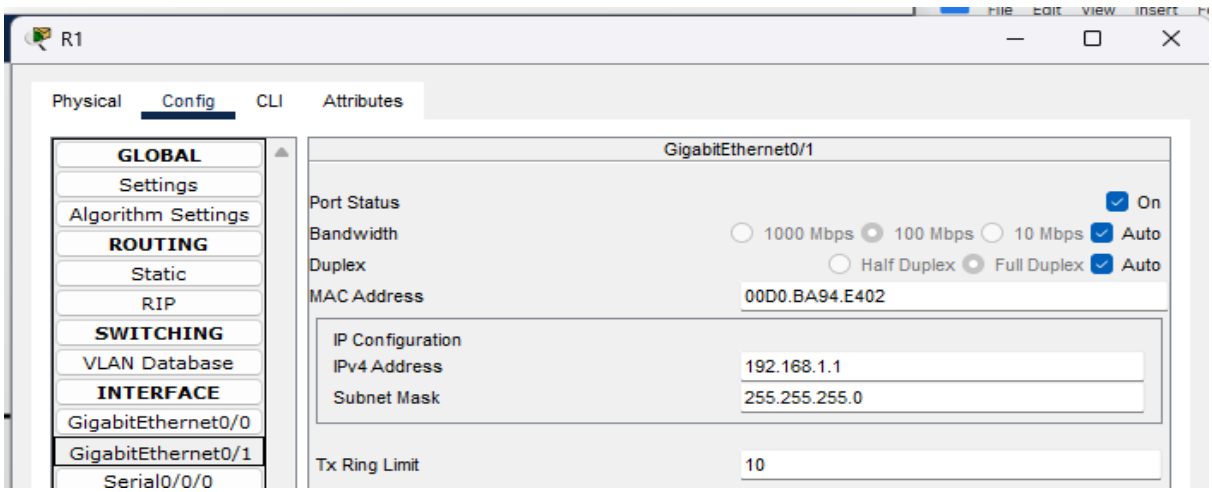


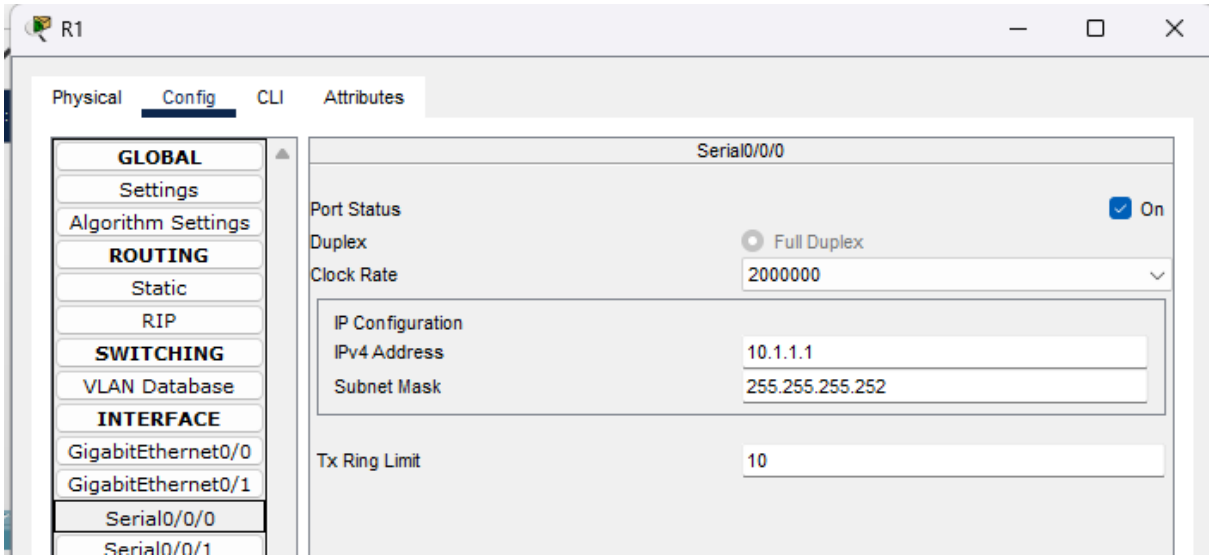
Configuration:

PC-A

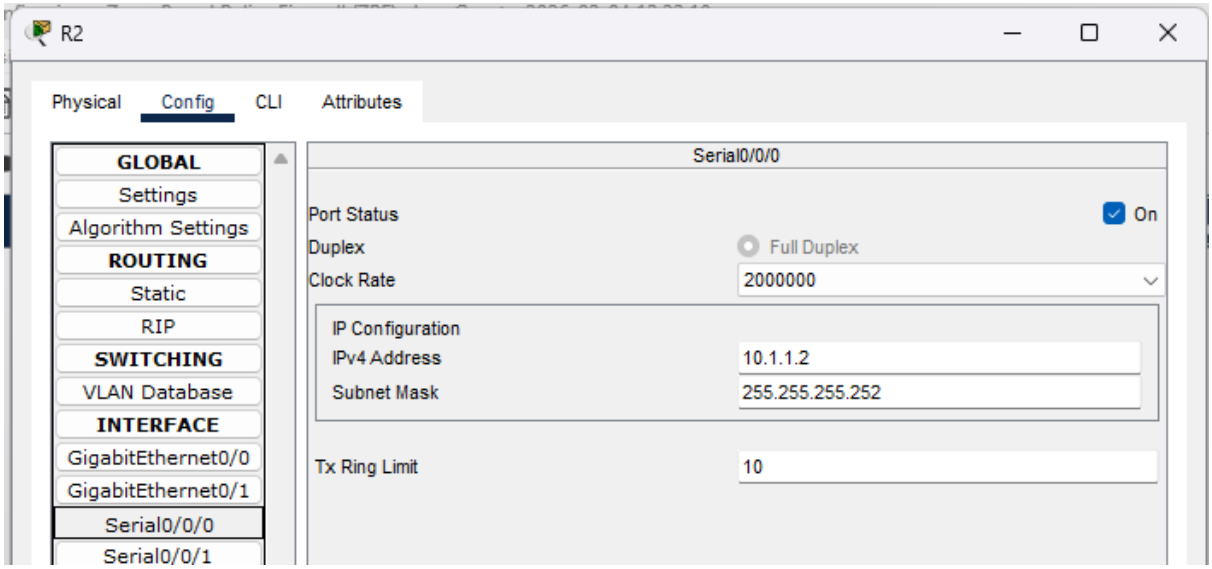


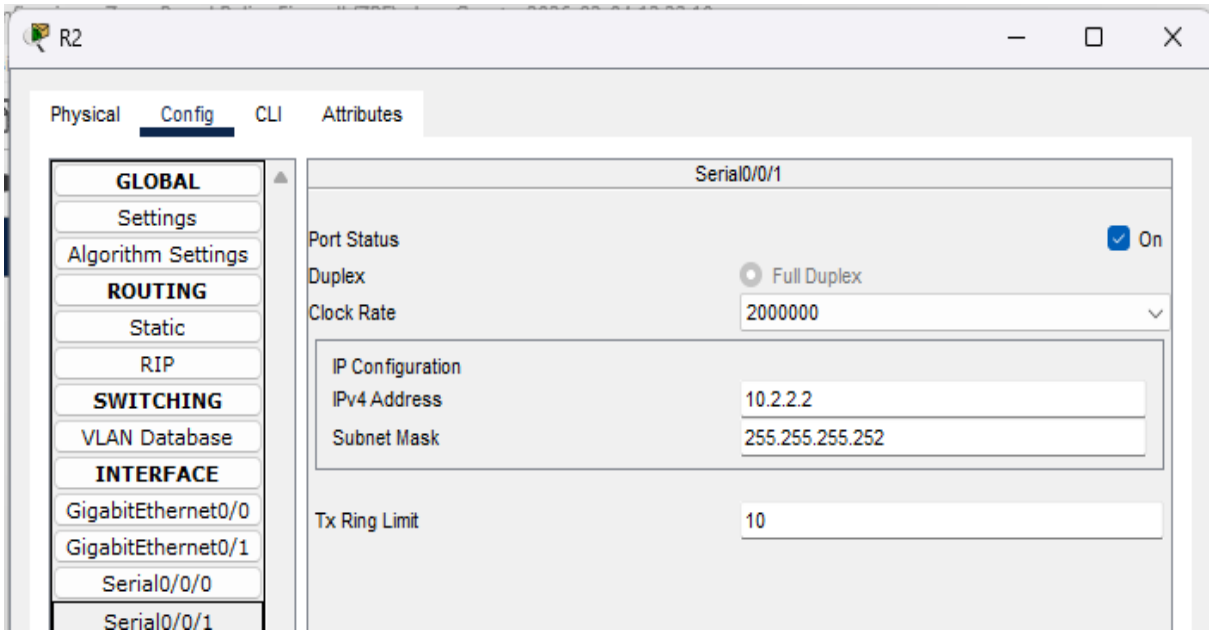
R1



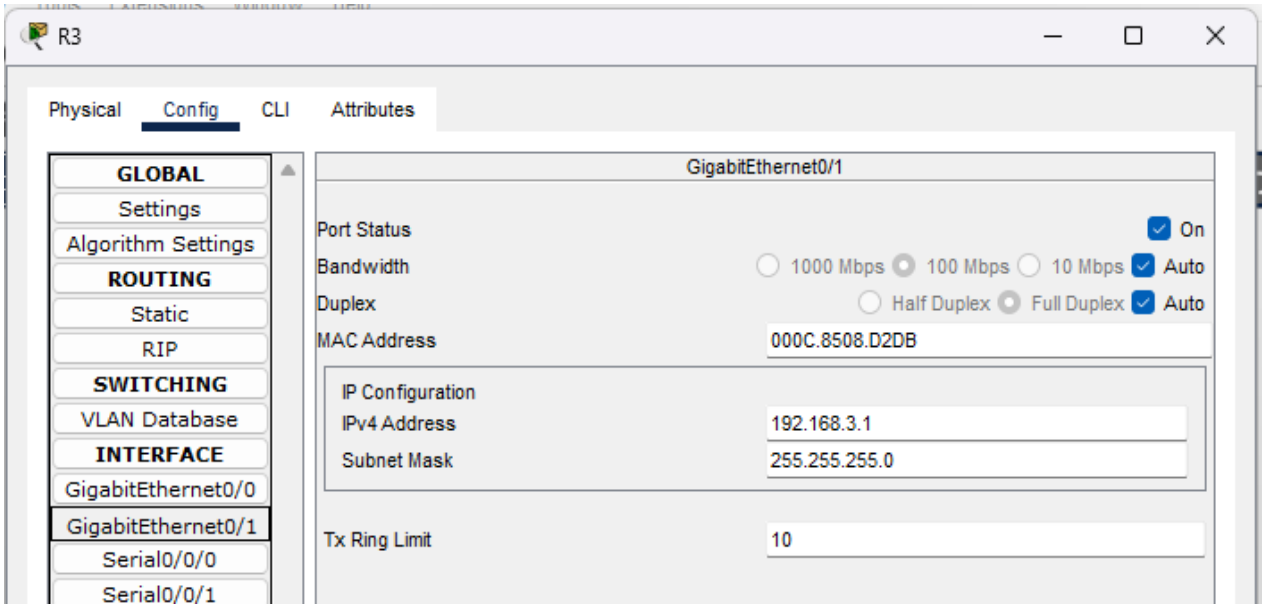

R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

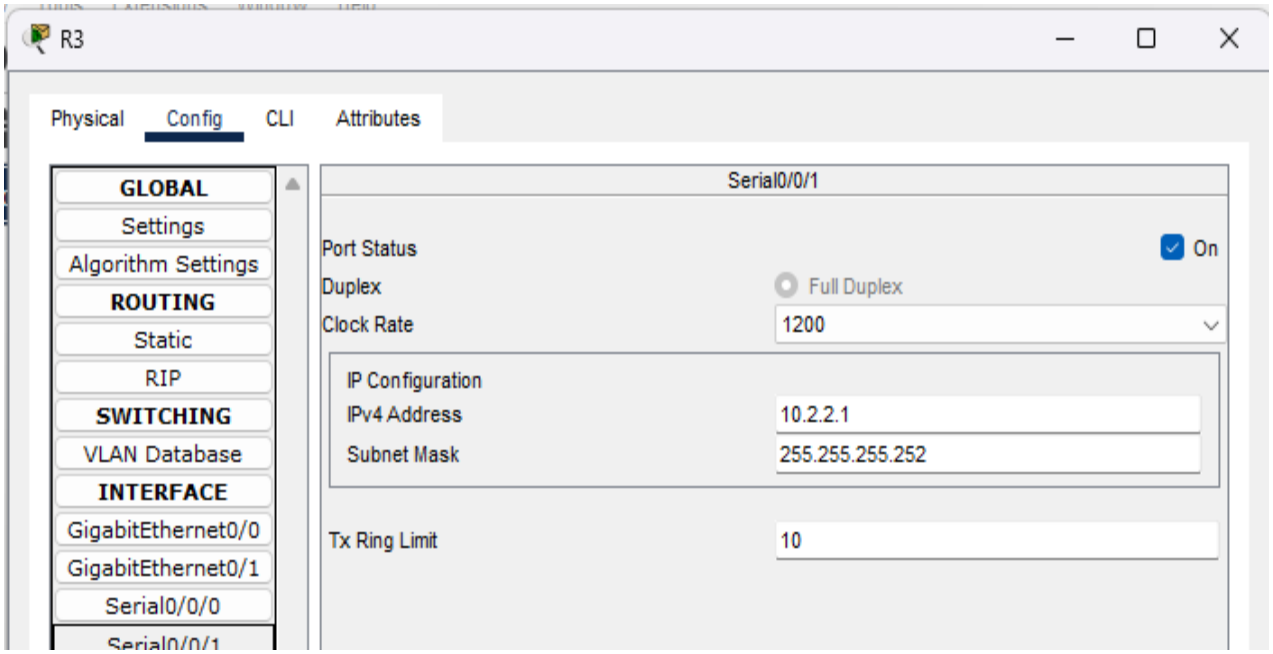
R2


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

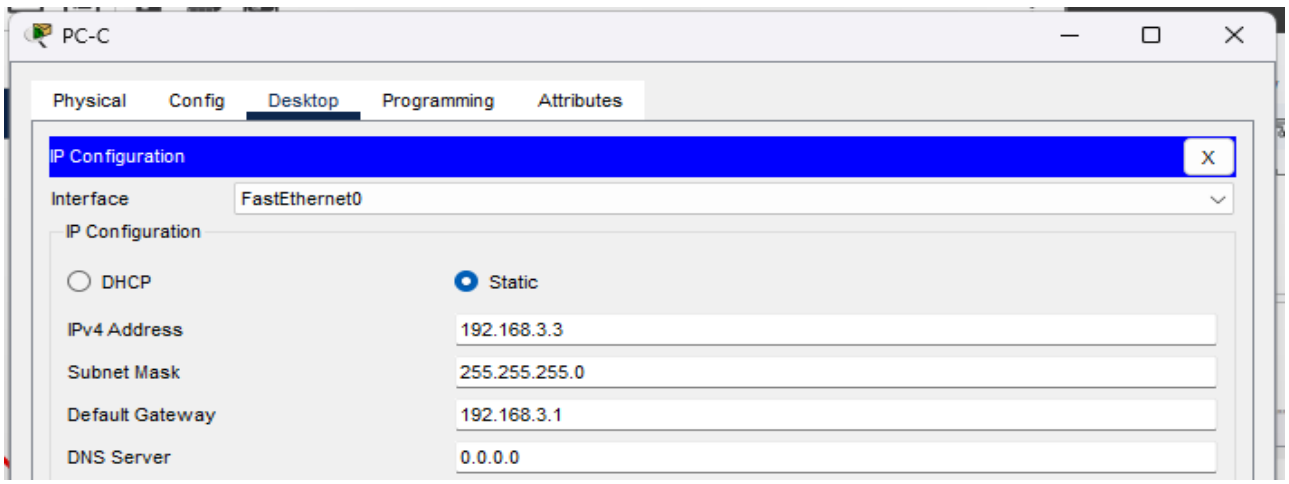

R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.2.2.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10

R3


R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000C.8508.D2DB
IP Configuration
IPv4 Address
192.168.3.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10


R3
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
1200
IP Configuration
IPv4 Address
10.2.2.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10

PC-C


PC-C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.3
Subnet Mask
255.255.255.0
Default Gateway
192.168.3.1
DNS Server
0.0.0.0

Part 1: Verify Basic Network Connectivity

PC-A

PC-A

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Reply from 192.168.3.3: bytes=32 time=8ms TTL=125
Reply from 192.168.3.3: bytes=32 time=7ms TTL=125
Reply from 192.168.3.3: bytes=32 time=10ms TTL=125
Reply from 192.168.3.3: bytes=32 time=7ms TTL=125

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 10ms, Average = 8ms

C:\>
```

R2

```
R2>en
Password:
Password:
Password:
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip domain-name ccnasecurity.com
R2(config)#enable secret ciscoenpa55
R2(config)#line console 0
R2(config-line)#exit
R2(config)#username admin secret adminpa55
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#exit
R2(config)#
```

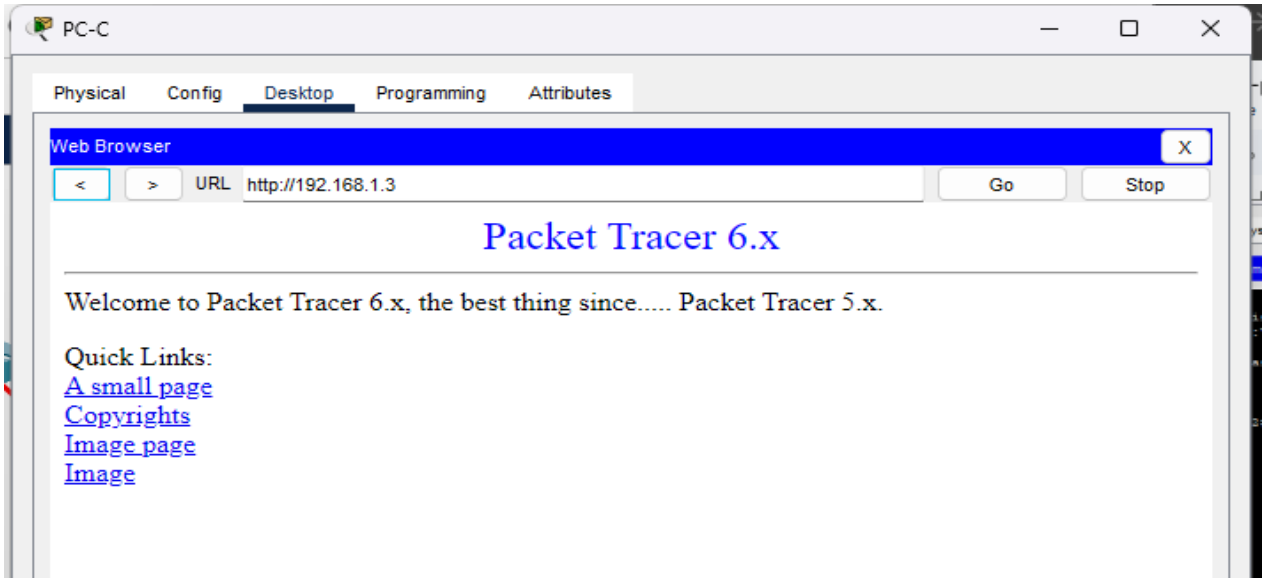
PC-C

PC-C

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 10.2.2.2

Password:



R2#
```



Part 2: Create the Firewall Zones on R3

R3

```
R3>en
Password:
R3#show version
Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE
SOFTWARE (fc2)
```

```
License Info:

License UDI:

-------------------------------------------------
Device#   PID                   SN
-------------------------------------------------
*0        CISCO1941/K9          FTX152446W3


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      disable       None          None
data          disable       None          None

Configuration register is 0x2102
```

```
R3(config)#license boot module c1900 technology-package securityk9
PLEASE  READ THE  FOLLOWING TERMS  CAREFULLY. INSTALLING THE LICENSE OR
LICENSE  KEY  PROVIDED FOR  ANY CISCO  PRODUCT  FEATURE  OR  USING SUCH
PRODUCT  FEATURE  CONSTITUTES  YOUR  FULL ACCEPTANCE  OF  THE FOLLOWING
```

```
R3#reload
System configuration has been modified. Save? [yes/no]:y
Building configuration...
[OK]
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
```

```
R3>en
Password:
R3#show version
```

```
License Info:

License UDI:

-------------------------------------------------
Device#   PID                    SN
-------------------------------------------------
*0        CISCO1941/K9           FTX152446W3


Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102


R3#
```

Part 3: Identify Traffic Using a Class-Map

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
R3#en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#zone security in-zone
R3(config-sec-zone)#exit
R3(config)#zone security out-zone
R3(config-sec-zone)#exit
R3(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 any
R3(config)#class-map type inspect match-all in-net-class-map
R3(config-cmap)#match access-group 101
R3(config-cmap)#exit
R3(config)#policy-map type inspect in-2-out-pmap
```

Part 4: Specify Firewall Policies

```
R3(config)#policy-map type inspect in-2-out-pmap
R3(config-pmap)#class type inspect in-net-class-map
R3(config-pmap-c)#inspect
%No specific protocol configured in class in-net-class-map for inspection. All protocols
will be inspected
R3(config-pmap-c)#exit
R3(config-pmap)#exit
```

Part 5: Apply Firewall Policies

```
R3(config)#zone-pair security in-2-out-zpair source in-zone destination out-zone
R3(config-sec-zone-pair)#service-policy type inspect in-2-out-pmap
R3(config-sec-zone-pair)#exi
R3(config)#interface g0/1
R3(config-if)#zone-member security in-zone
R3(config-if)#exit
R3(config)#interface s0/0/1
R3(config-if)#zone-member security out-zone
R3(config-if)#exit
R3(config)#exit
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#copy running-config startup-config
```

Part 6: Test Firewall Functionality from IN-ZONE to OUT-ZONE

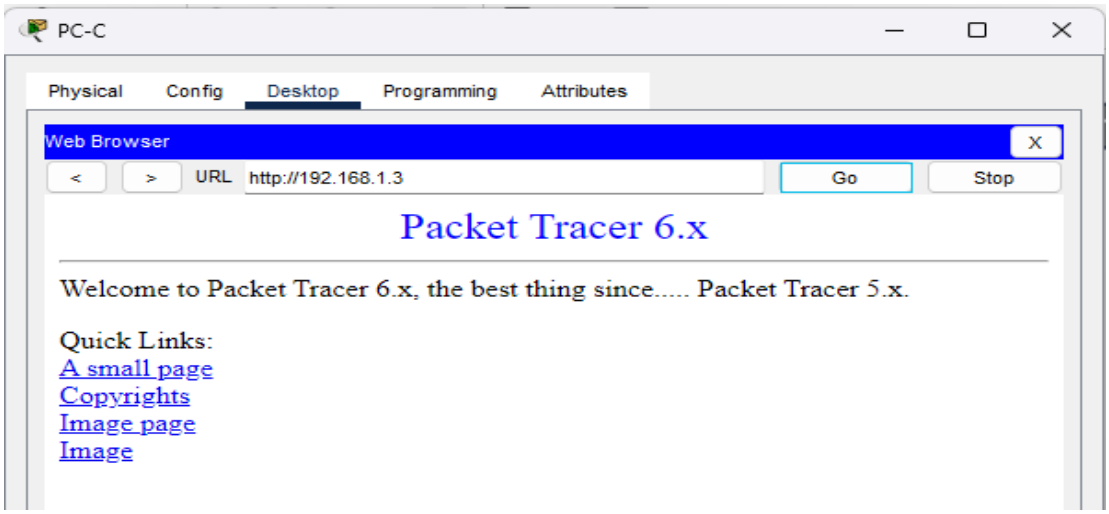
PC-C

```
PC-C
Physical  Config  Desktop  Programming  Attributes
Command Prompt

[Connection to 10.2.2.2 closed by foreign host]
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.3: bytes=32 time=9ms TTL=125
Reply from 192.168.1.3: bytes=32 time=10ms TTL=125
Reply from 192.168.1.3: bytes=32 time=7ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 10ms, Average = 8ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=10ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 4ms

C:\>ssh -l admin 10.2.2.2

Password:


R2#
```

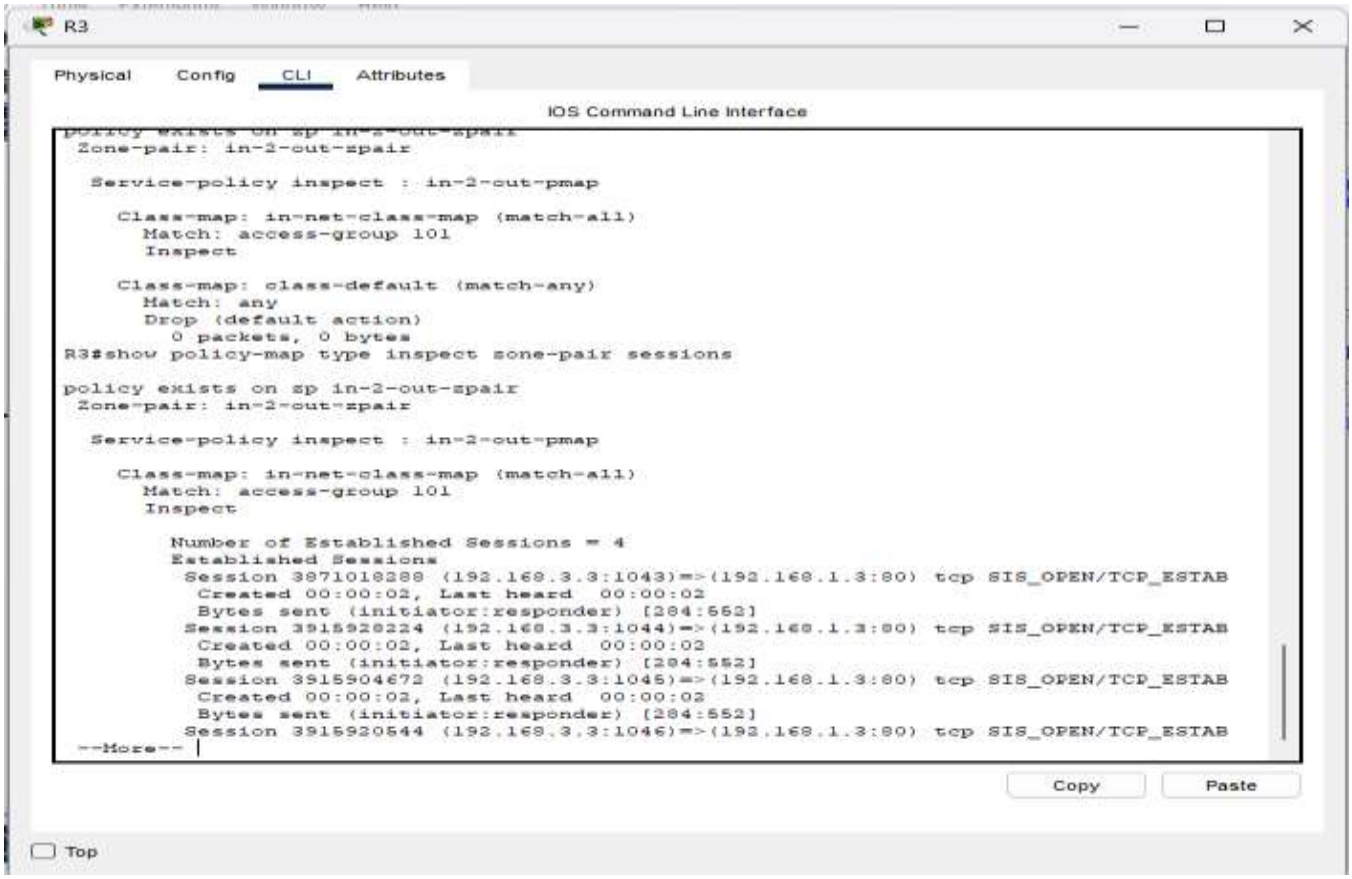
R3

```
R3#
R3#show policy-map type inspect zone-pair sessions

policy exists on zp in-2-out-zpair
 Zone-pair: in-2-out-zpair

  Service-policy inspect : in-2-out-pmap

    Class-map: in-net-class-map (match-all)
      Match: access-group 101
      Inspect

    Class-map: class-default (match-any)
      Match: any
      Drop (default action)
        0 packets, 0 bytes
R3#
```

PC-C

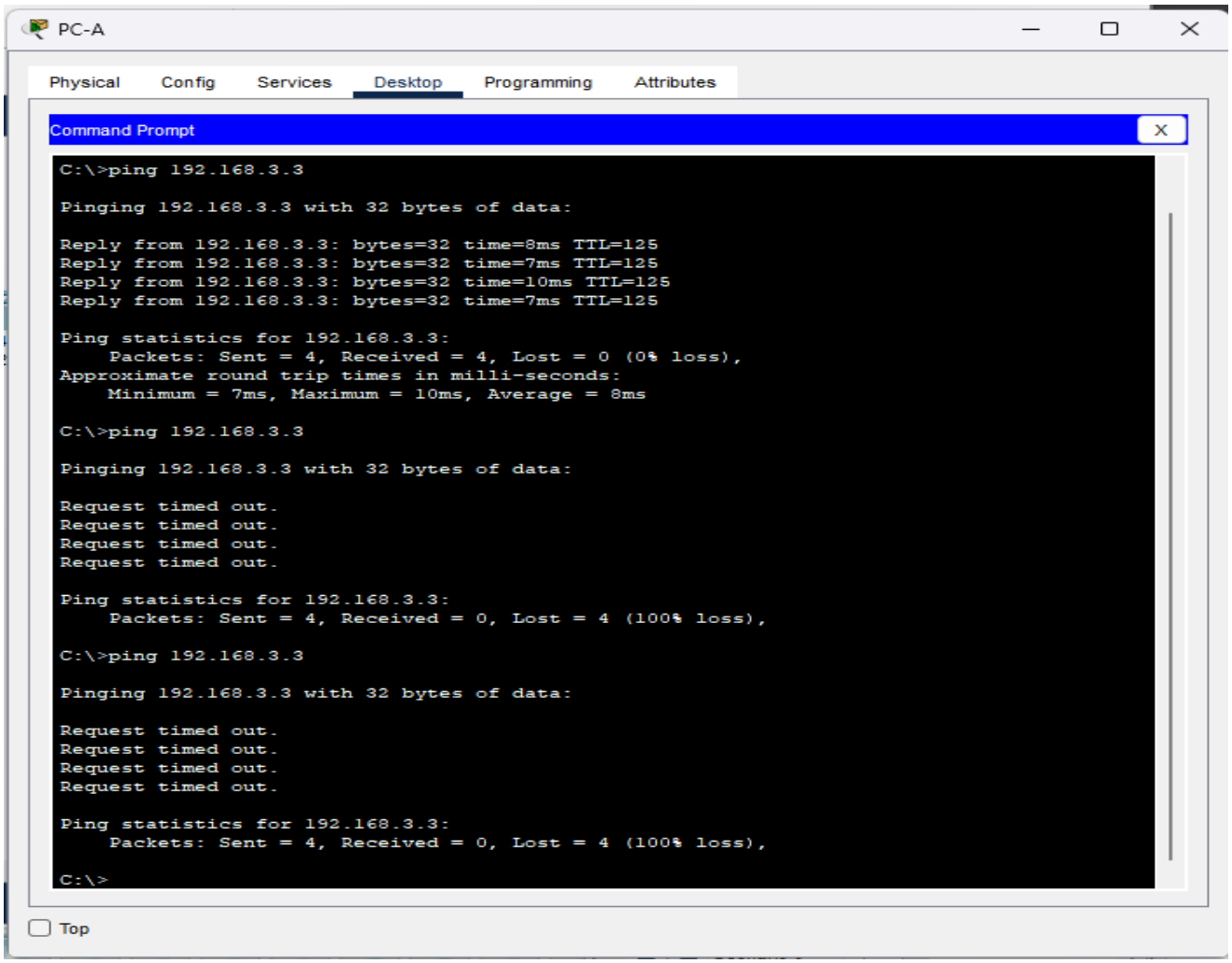


Run the command immediately after searching on the web browser



Part 7: Test Firewall Functionality from OUT-ZONE to IN-ZONE

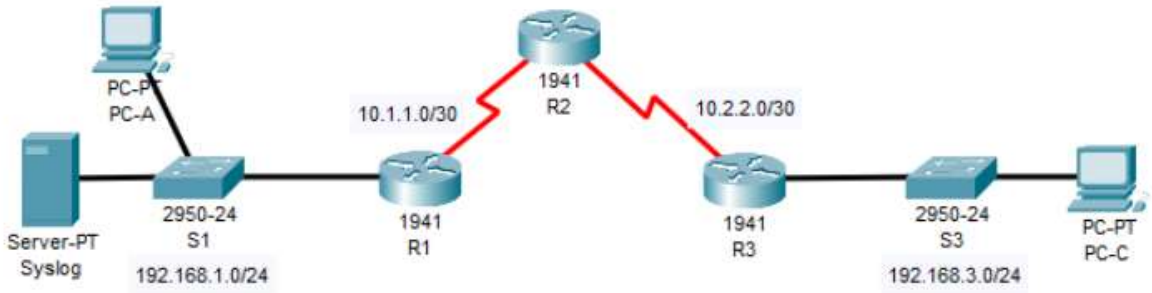
PC-A


PC-A
Physical
Config
Services
Desktop
Programming
Attributes
Command Prompt
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Reply from 192.168.3.3: bytes=32 time=8ms TTL=125
Reply from 192.168.3.3: bytes=32 time=7ms TTL=125
Reply from 192.168.3.3: bytes=32 time=10ms TTL=125
Reply from 192.168.3.3: bytes=32 time=7ms TTL=125
Ping statistics for 192.168.3.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 7ms, Maximum = 10ms, Average = 8ms
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.3.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.3.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top

Practical - 6

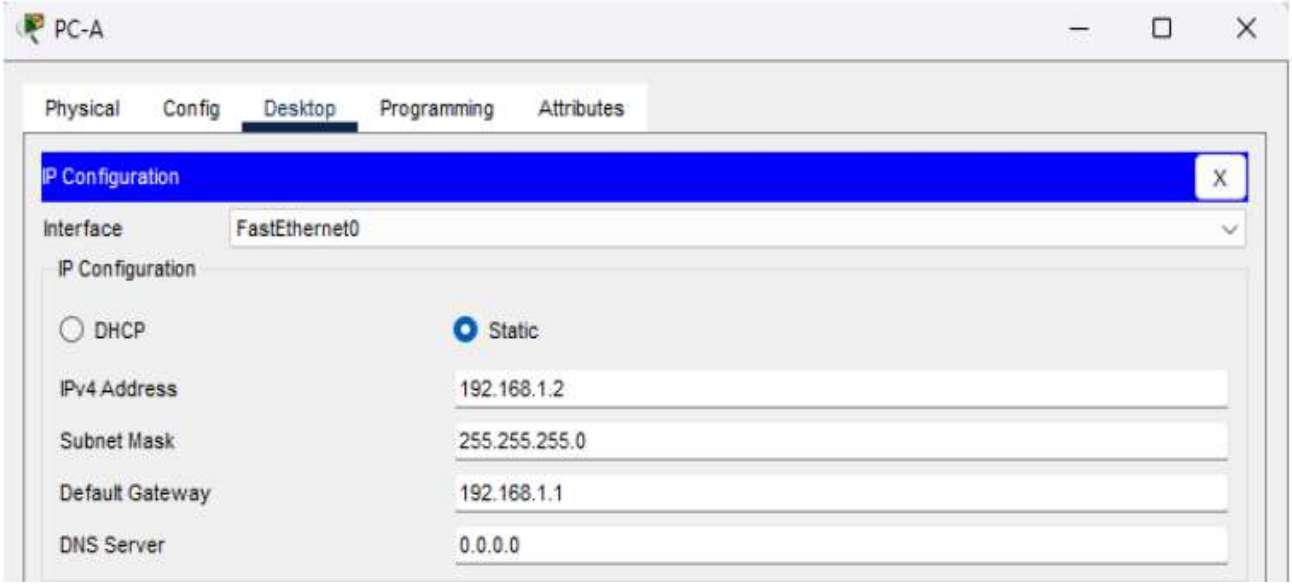
Configure IOS Intrusion Prevention System (IPS) Using the CLI

Topology:

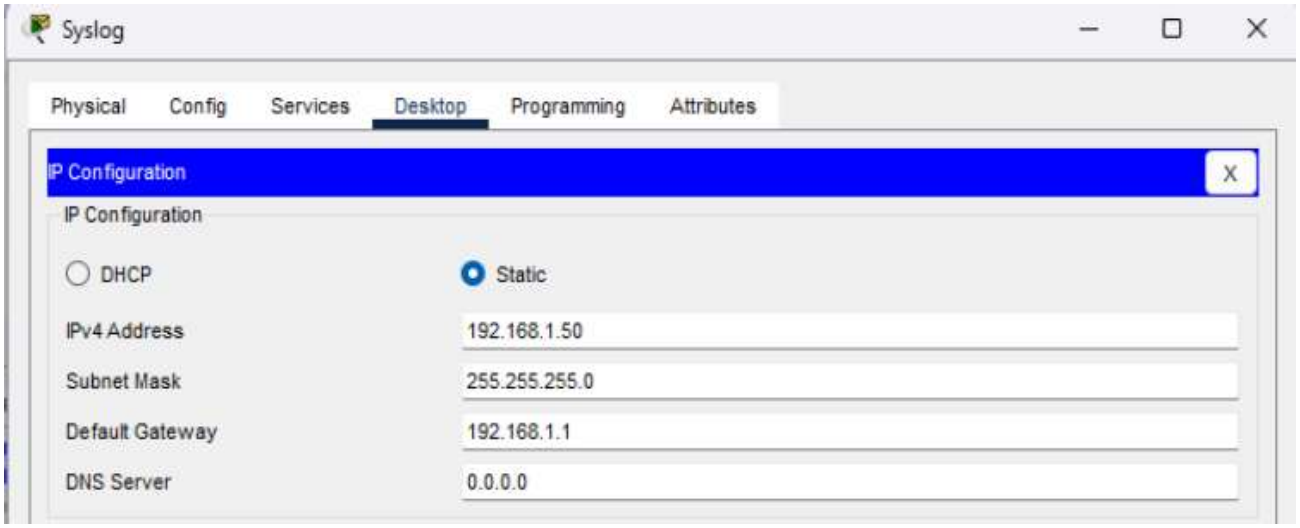


Configurations:

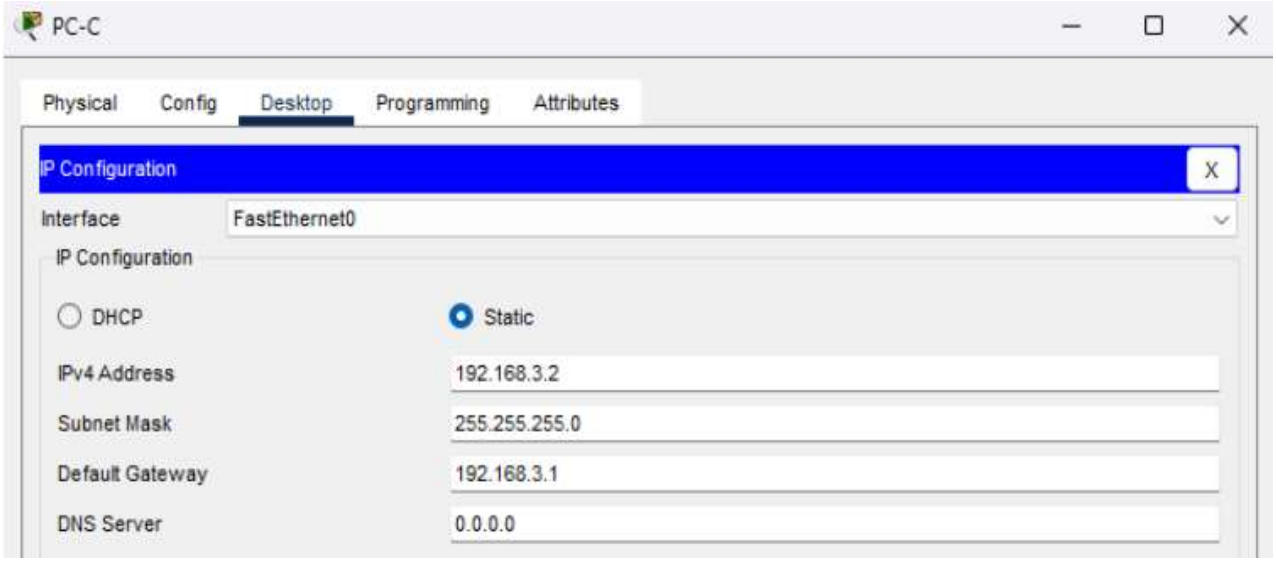
PC-A



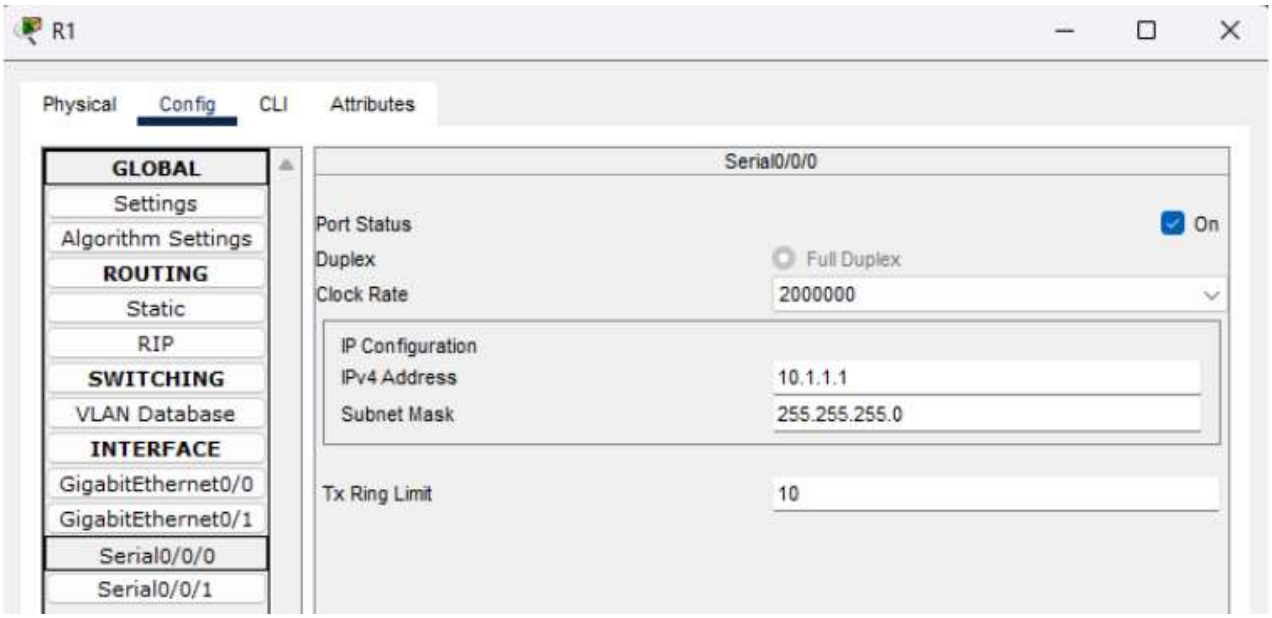
Syslog

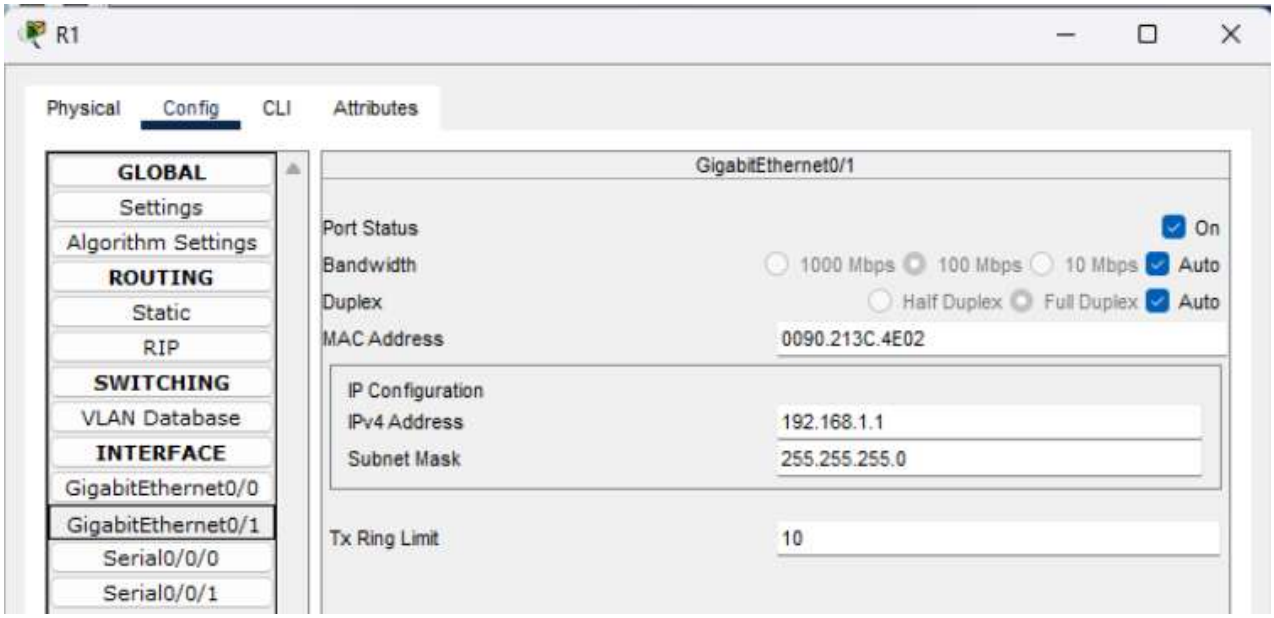


PC-C

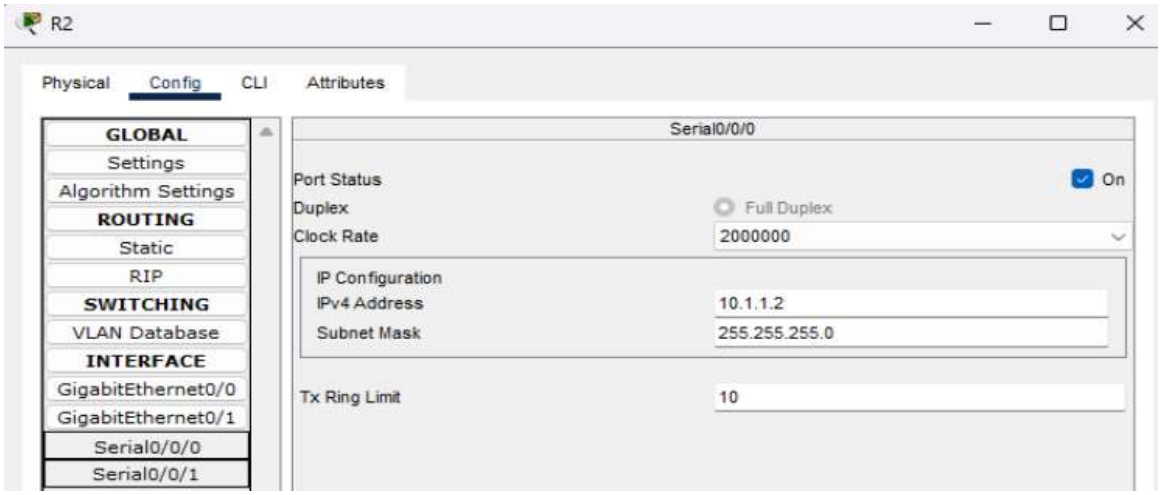

PC-C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.3.1
DNS Server
0.0.0.0

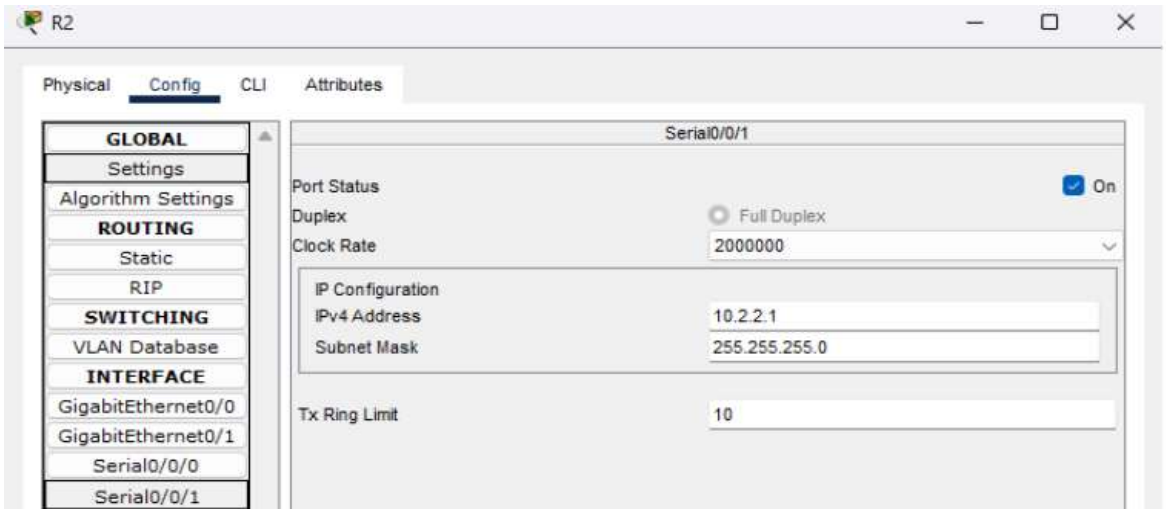
R1


R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

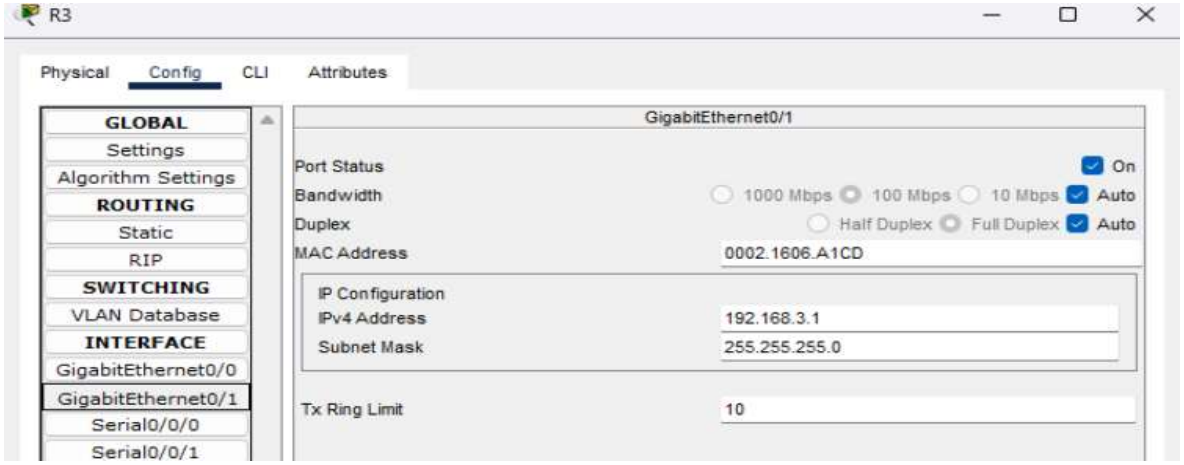

R1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.213C.4E02
IP Configuration
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

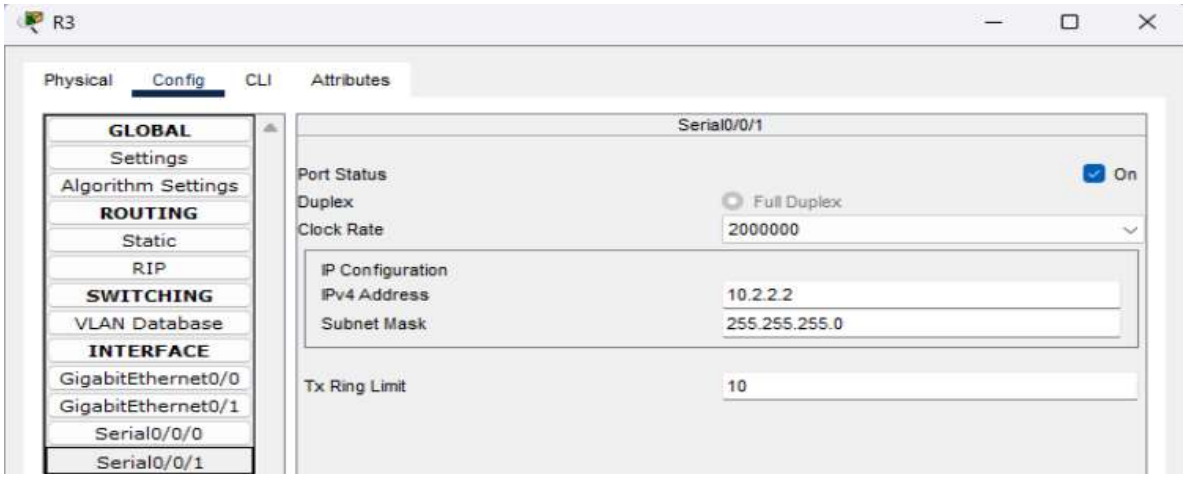
R2


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.1.1.2
Subnet Mask
255.255.255.0
Tx Ring Limit
10


R2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/1
Port Status
On
Duplex
Full Duplex
Clock Rate
2000000
IP Configuration
IPv4 Address
10.2.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

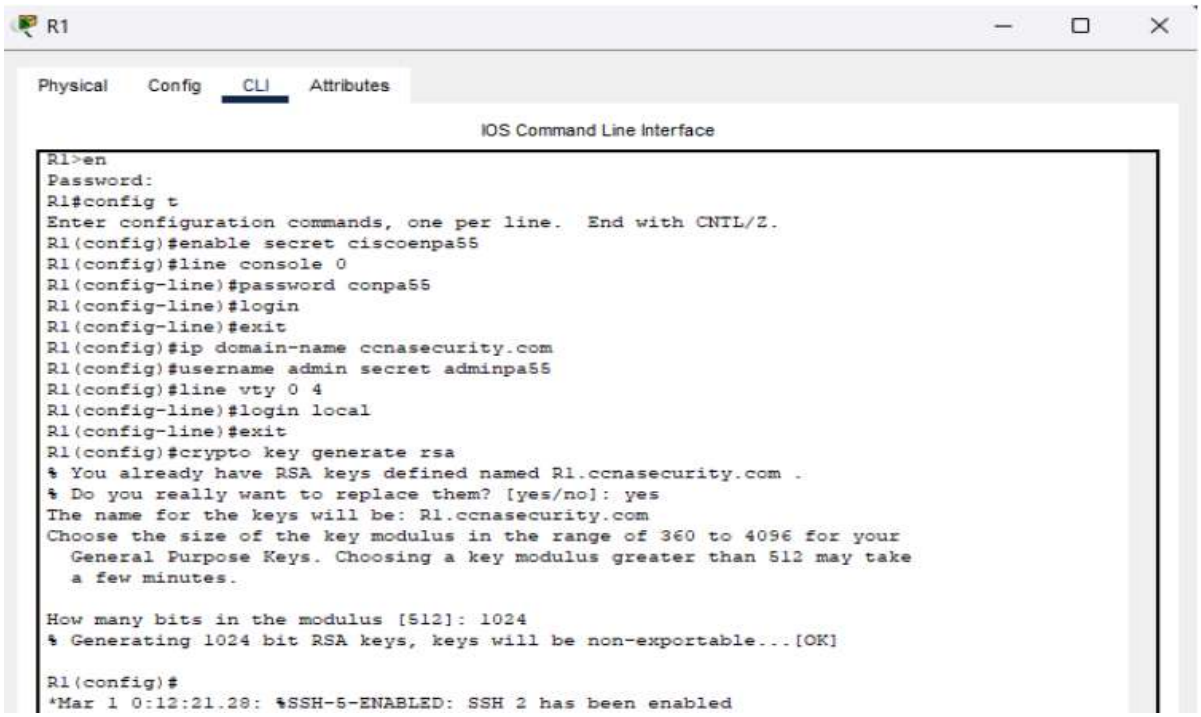
R3





Part 1: Enable IOS IPS

R1 CLI



PC-A -> PC-C

PC-A

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=7ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=10ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 5ms

C:\>
```

PC-C -> PC-A

PC-C

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=10ms TTL=125
Reply from 192.168.1.2: bytes=32 time=10ms TTL=125
Reply from 192.168.1.2: bytes=32 time=8ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 7ms

C:\>
```

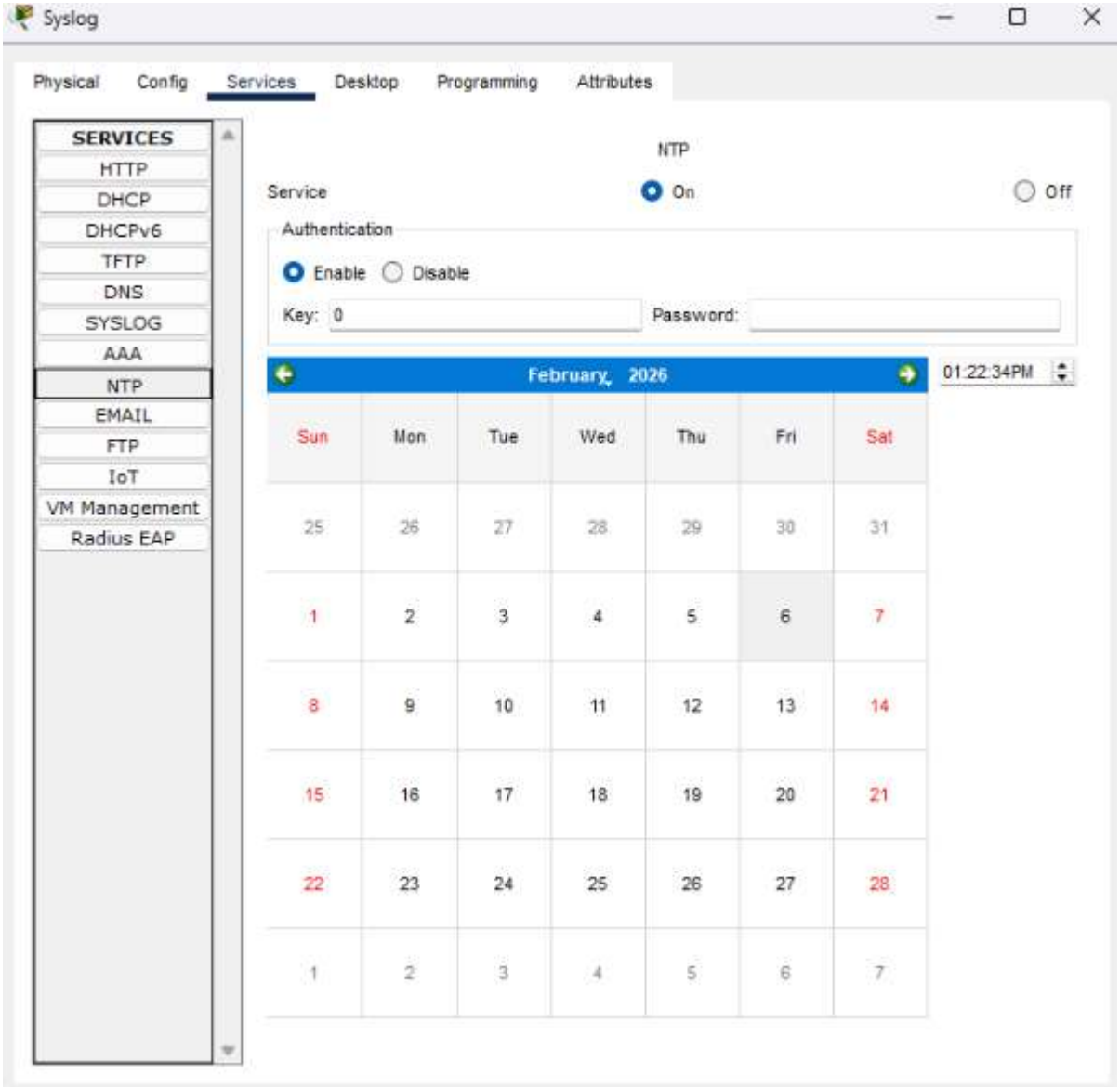
R1 CLI

```
R1(config)#license boot module c1900 technology-package securityk9
```

```
Technology Package License Information for Module:'c1900'

-----------------------------------------------------------------
Technology    Technology-package          Technology-package
              Current       Type          Next reboot
-----------------------------------------------------------------
ipbase        ipbasek9      Permanent     ipbasek9
security      securityk9    Evaluation    securityk9
data          disable       None          None

Configuration register is 0x2102
```

```
R1#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir

R1#
```

Syslog


Syslog
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
NTP
Service
On
Off
Authentication
Enable
Disable
Key: 0
Password:
February, 2026
01:22:34PM
Sun Mon Tue Wed Thu Fri Sat

R1

R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
R1(config)#ip ips config location flash:ipsdir
R1(config)#ip ips name iosips
R1(config)#ip ips notify log
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#clock set 10:18:00 6 February 2026
R1#

```
R1(config)#service timestamps log datetime msec
R1(config)#logging host 192.168.1.50
R1(config)#ip ips signature-category
R1(config-ips-category)#category all
R1(config-ips-category-action)#retired true
R1(config-ips-category-action)#exit
R1(config-ips-category)#category ios_ips basic
R1(config-ips-category-action)#retired false
R1(config-ips-category-action)#exit
R1(config-ips-category)#exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this engine will be
scanned

R1(config)#
```

```
R1(config)#interface g0/1
R1(config-if)#ip ips iosips out
R1(config-if)#
*Feb 06, 10:23:44.2323:  %IPS-6-ENGINE_BUILDS_STARTED:  10:23:44 UTC Feb 06 2026
*Feb 06, 10:23:44.2323:  %IPS-6-ENGINE_BUILDING: atomic-ip - 3 signatures - 1 of 13
engines
*Feb 06, 10:23:44.2323:  %IPS-6-ENGINE_READY: atomic-ip - build time 8 ms - packets for
this engine will be scanned
*Feb 06, 10:23:44.2323:  %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 8 ms
R1(config-if)#
```

Part 2: Modify the Signature

R1 CLI

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1(config)#ip ips signature-definition
R1(config-sigdef)#signature 2004 0
R1(config-sigdef-sig)#status
R1(config-sigdef-sig-status)#retired false
R1(config-sigdef-sig-status)#enabled true
R1(config-sigdef-sig-status)#exit
R1(config-sigdef-sig)#engine
R1(config-sigdef-sig-engine)#event-action produce-alert
R1(config-sigdef-sig-engine)#event-action deny-packet-inline
R1(config-sigdef-sig-engine)#exit
R1(config-sigdef-sig)#exit
R1(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this engine will be
scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms
```

```
R1#show ip ips all
IPS Signature File Configuration Status
    Configured Config Locations: flash:ipsdir
    Last signature default load time:
    Last signature delta load time:
    Last event action (SEAP) load time: -none-

    General SEAP Config:
    Global Deny Timeout: 3600 seconds
    Global Overrides Status: Enabled
    Global Filters Status: Enabled

IPS Auto Update is not currently configured

IPS Syslog and SDEE Notification Status
    Event notification through syslog is enabled
    Event notification through SDEE is enabled

IPS Signature Status
    Total Active Signatures: 1
    Total Inactive Signatures: 0

IPS Packet Scanning and Interface Status
    IPS Rule Configuration
      IPS name iosips
    IPS fail closed is disabled
    IPS deny-action ips-interface is false
    Fastpath ips is enabled
    Quick run mode is enabled
    Interface Configuration
      Interface GigabitEthernet0/1
        Inbound IPS rule is not set
        Outgoing IPS rule is iosips
```

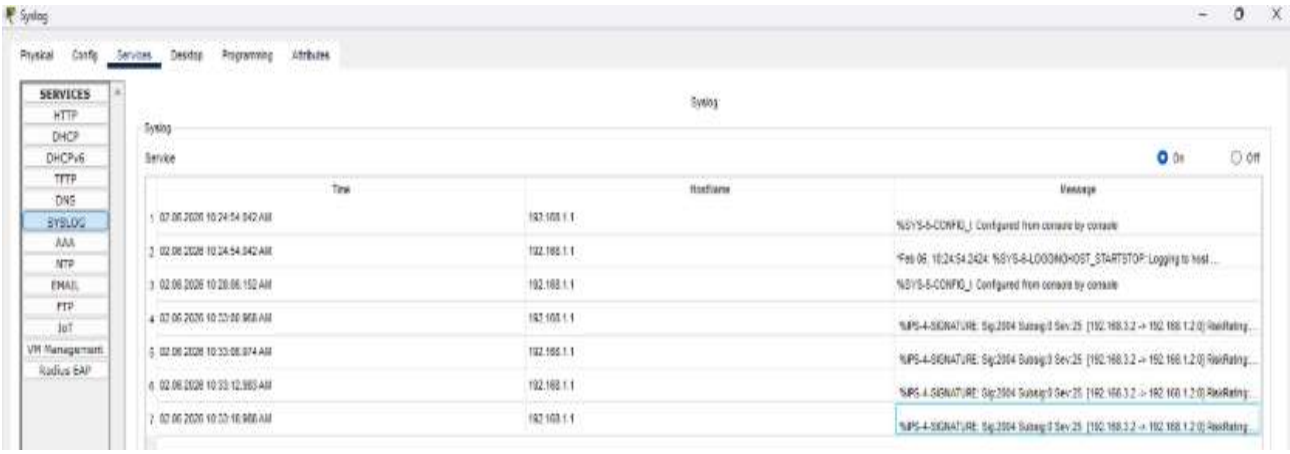
PC-A -> PC-C

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=4ms TTL=125
Reply from 192.168.3.2: bytes=32 time=6ms TTL=125
Reply from 192.168.3.2: bytes=32 time=8ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 8ms, Average = 5ms
```

PC-C -> PC-A

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Syslog


Syslog
Physical
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SERVICES
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DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
Syslog
Syslog
Service
On
Off
Time
Hostname
Message
192.168.1.1
%SYS-5-CONFIG_I: Configured from console by console
192.168.1.1
*Feb 06, 10:24:54.2424: %SYS-6-LOGGMO=DST_STARTSTOP: Logging to host...
192.168.1.1
%SYS-5-CONFIG_I: Configured from console by console
192.168.1.1
192.168.1.1
192.168.1.1
192.168.1.1